# Harmonic Branching and Quantitative Sieve Bounds for Mills-Type Prime-Representing Functions

An explicit conditional exponent below 24/23

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**Abstract**

For $c > 1$, let $\mathcal{W}(c)$ be the set of real numbers $A > 1$ for which $\lfloor A^{c^n} \rfloor$ is prime for every $n \geq 1$. We prove a harmonic mass–compactness principle that converts a quantitative almost-all prime lower bound into uncountably many representing constants. An exceptional measure of size $X \exp[-D(\log\log X)^2]$, with sufficiently large fixed $D$, controls all depths of the prime tree simultaneously. The argument also treats bounded variable exponent sequences and proves unboundedness, Lebesgue nullity, and nowhere density. It recovers the zero-density ranges $c > 6/5$ and $c > 15/13$, and quantifies the finite sieve inputs leading to $c > 20/19$ and $c > 43/41$ with their respective source dependencies. For the sharper extension, we identify an internal integer-moment obstruction at the short-interval exponent $1/24$ and remove it by a two-factor estimate valid at $1/25$, using the Chen–Gupta–Li large-value theorem. Exact rational certificates verify the additional trilinear regions and the finite boundary geometry. Assuming four explicitly stated numerical Buchstab inequalities from Li's third short-interval manuscript, we obtain $\theta_* = 1/24 - 10^{-23}$ and hence all $c > 24/(23 + 24 \cdot 10^{-23})$. In particular, $c = 24/23 - 10^{-23}$ is admissible. The four numerical inequalities remain external hypotheses; the exploratory recomputation of one integral is not an interval certificate. The full prime-interval exponent $1/25$ is not claimed. The single source embeds the exact verification programs and certificates.

**Keywords:** prime-representing functions; Mills exponent; harmonic mass; primes in short intervals; Buchstab decomposition; Dirichlet polynomials; exact rational certificates.
**MSC 2020:** 11N05, 11N35, 11M26.

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# 1 Introduction and principal results

Mills' prime-representing construction asks for a real number whose successive prescribed powers have prime integer parts [1]. Write $\mathcal{P}$ for the primes and define

$$\mathcal{W}(c) = \{A > 1 : \lfloor A^{c^n} \rfloor \in \mathcal{P} \text{ for every } n \geq 1\}.$$

Here $c$ is the exponent parameter, and $A$ is the representing constant. An improvement in $c$ does not compute or alter the least usual Mills constant associated with exponent 3. For a sequence $\boldsymbol{c} = (c_n)_{n\geq 1}$, put $C_n = c_1 \cdots c_n$ and

$$\mathcal{W}(\boldsymbol{c}) = \{A > 1 : \lfloor A^{C_n} \rfloor \in \mathcal{P} \text{ for every } n \geq 1\}.$$

The nested prime cells are $J_c(m) = [m^c, (m+1)^c - 1)$. At height $X \asymp m^c$, their lengths are comparable to $X^{1-1/c}$. A prime lower bound in almost all intervals of length $X^\theta$ is therefore relevant when $\theta < 1 - 1/c$. The remaining difficulty is to keep prime-rich cells at every depth. The harmonic construction measures the entire finite level by $\sum_p 1/p$. It compares its accumulated mass with the mass of all defective integers at that scale, and then uses finite branching compactness. The full argument, including the uniformity in depth, appears in Section 4.

**Theorem 1.1** (explicit conditional Mills range)**.** *Assume the four numerical inequalities of Hypothesis 1.3, with the literal printed Li III regions and the conventions specified there. Let*

$$\Delta = 10^{-23}, \qquad \theta_* = \frac{1}{24} - \Delta, \qquad c_* = \frac{1}{1-\theta_*} = \frac{24}{23 + 24\Delta}.$$

*For every fixed $D > 0$, there is $\kappa > 0$ such that, for sufficiently large $X$,*

$$\operatorname{meas}\left\{x \in [X, 2X] : \pi(x) - \pi(x - X^{\theta_*}) < \kappa \frac{X^{\theta_*}}{\log X}\right\} \ll X \exp[-D(\log\log X)^2]. \tag{1}$$

*For every sequence satisfying $c_* < \inf_n c_n \leq \sup_n c_n < \infty$, the set $\mathcal{W}(\boldsymbol{c})$ is uncountable, unbounded, Lebesgue-null, and nowhere dense. In particular, this holds for every constant $c > c_*$, including the exact rational value*

$$\boxed{c_{\text{new}} = \frac{24}{23} - 10^{-23}}.$$

The lower boundary $c_*$ is strict. The displayed $c_{\text{new}}$ is a convenient admissible point, since

$$c_{\text{new}} - c_* = \frac{\Delta(47 - 552\Delta)}{23(23 + 24\Delta)} > 0. \tag{2}$$

The endpoint $c = 24/23$ is also included. No optimality is asserted for $\Delta$, and the theorem does not include $c = c_*$.

**Theorem 1.2** (zero-density specializations)**.** *The conclusions about $\mathcal{W}(\boldsymbol{c})$ hold unconditionally when $6/5 < \inf_n c_n \leq \sup_n c_n < \infty$, using the classical Ingham–Huxley zero-density estimates. With the Guth–Maynard estimate, the lower bound $6/5$ may be replaced by $15/13$. Under the density hypothesis $N(\sigma, T) \ll_\varepsilon T^{2(1-\sigma)+\varepsilon}$, it may be replaced by $1$.*

Section 5 proves these specializations without the four numerical hypotheses used in Theorem 1.1. The intermediate sieve ranges are retained in Section 8. Their numerical inputs and those of the new theorem are separate.

Table 1: Inputs and the ranges established by the corresponding arguments. The inequalities in the Mills column are strict.

| Input | Prime-interval scale | Mills range | Dependency |
|---|---|---|---|
| Ingham–Huxley | $\theta > 1/6$ | $c > 6/5$ | Published density estimates |
| Guth–Maynard | $\theta > 2/15$ | $c > 15/13$ | Published density estimate |
| Jia finite sieve | $\theta > 1/20$ | $c > 20/19$ | Quantified source proof and its published numerical margin |
| Li I finite sieve | $\theta > 2/43$ | $c > 43/41$ | Its three stated numerical region bounds |
| Present extension | $\theta_* = 1/24 - 10^{-23}$ | $c > c_*$ | Four Li III bounds; CGL large values; exact certificates |
| Density hypothesis | Every $\theta > 0$ | Every $c > 1$ | Conditional zero-density assertion |

**Proof dependencies.** The harmonic transfer is proved independently of the sieve. The analytic extension first repairs the two-factor obstruction, then verifies the trilinear and Type I regions, and finally stabilizes a finite signed Buchstab decomposition. The exact comparator-loss bound is 0.00295856, against an assumed original margin greater than 0.02. The positive margin supplies (1); harmonic transfer then proves the Mills assertion. The two-factor lemma itself works at 1/25, but no positive final sieve margin at 1/25 is established.

The Chen–Gupta–Li result [2] and Li III [3] are used at the stated preprint/manuscript versions. The numerical hypotheses are not replaced by the exact finite-geometry checks. This consolidated paper contains the earlier harmonic and quantitative arguments, so none of its proofs requires an unpublished predecessor by the author. It makes no claim of priority over all other work.

### 1.1 The precise numerical sieve hypothesis

The numerical input is stated here as four definite inequalities. Its regions are the literal regions in Li [3, Section 7, equations (15), (23), (24), and (26)]; in particular the printed empty region $V_0$ is not silently replaced by a larger one. Boundary conventions on finitely many hyperplanes do not affect the integrals below.

Put $a = 11/126$, $\theta_0 = 1/24$, $s = t_1 + t_2$, and

$$\Omega = \{(t_1, t_2) : a \le t_1 < 1/2, \quad a \le t_2 < \min(t_1, (1 - t_1)/2)\}.$$

Using the seven raw regions $\mathcal{C}_j(\theta_0)$ recorded in Appendix A, define

$$U_{01} = \Omega \cap \bigcup_{j=1}^{7} \mathcal{C}_j(\theta_0).$$

The three printed additional components are

$$\begin{aligned} V_0 &= \{(t_1, t_2) : t_2 > 1/12, \quad 2t_2 - 1/12 < t_1 < 1/16\}, \\ V_{11} &= \{(t_1, t_2) : |4t_1 + 3t_2 - 2| < 1/12, \quad 1/12 < t_2 < 1/6\}, \\ V_{12} &= \{(t_1, t_2) : |4t_1 + t_2 - 2| < 1/12, \quad 1/12 < t_2 < 1/6\}, \\ V &= \Omega \cap (V_0 \cup V_{11} \cup V_{12}). \end{aligned}$$

Here $\Omega \cap V_0 = \varnothing$, since $a > 1/16$. The final displayed component is the second occurrence of the label $V_{11}$ in the source and is called $V_{12}$ here, consistently with the source's preceding union. This convention is fixed throughout the perturbation.

For $1 \le i \le 9$, let

$$B_i = \{(t_1, t_2) \in \Omega : \ell_i \le s \le u_i, \quad t_2 \le q_i s + r_i\},$$

where the exact endpoints are

| $i$ | $\ell_i$ | $u_i$ | $q_i$ | $r_i$ |
|---|---|---|---|---|
| 1 | 139/336 | 605/1456 | $-1$ | 29/52 |
| 2 | 605/1456 | 17/40 | $-41/27$ | 167/216 |
| 3 | 17/40 | 51/112 | $-4/3$ | 25/36 |
| 4 | 461/1008 | 11/24 | $-1/6$ | 35/144 |
| 5 | 11/24 | 23/48 | 0 | 1/6 |
| 6 | 23/48 | 265/552 | $-4$ | 25/12 |
| 7 | 265/552 | 4727/9816 | $-1/6$ | 35/144 |
| 8 | 4727/9816 | 151/312 | $-70/11$ | 71/22 |
| 9 | 151/312 | 503/1008 | $-4$ | 25/12 |

For $j = 1, 2, 3$, put

$$T_j = \{(t_1, t_2) \in \Omega : (t_1, t_2 + u) \in \mathcal{C}_j(\theta_0) \text{ for every } u \in [a, t_2]\}.$$

Because all defining inequalities are affine in $u$, testing its two endpoints gives exactly the source components $U_{0210}$, $U_{0211}$, and $U_{0212}$, respectively. Define

$$\widetilde{U}_{02} = \bigcup_{i=1}^{9} B_i \ \cup \ T_1 \cup T_2 \cup T_3.$$

For the remaining components use the two fixed Type I regions

$$W_1 = \{(m, h) : 2m + h \le 25/24, \quad 4m + 6h \le 73/24, \quad 4h \le 9/8\},$$
$$W_2 = \{(m, h) : m \le 25/48, \quad h \le 23/192\}.$$

Set

$$\begin{aligned}\widetilde{U}_{03} = \Omega \cap \big(&\{(t_1, t_2) : (s, t_2) \in W_1\} \cup \{(t_1, t_2) : (2t_2, t_1) \in W_1\} \\ &\cup \{(t_1, t_2) : 115/264 \le s \le 11/24, \quad t_2 \le 35/144 - s/6\}\big),\end{aligned}$$
$$\widetilde{U}_{04} = \Omega \cap \bigcup_{j=1}^{2} \{(t_1, t_2) : (t_1, t_2) \in W_j, \ (1 - s, t_2) \in W_j\}.$$

The disjoint priority domains used in the signed decomposition are

$$\begin{aligned}U_{02} &= \widetilde{U}_{02} \setminus (U_{01} \cup V), \\ U_{03} &= \widetilde{U}_{03} \setminus (U_{01} \cup V \cup U_{02}), \\ U_{04} &= \widetilde{U}_{04} \setminus (U_{01} \cup V \cup U_{02} \cup U_{03}), \\ U_0 &= U_{01} \cup V \cup U_{02} \cup U_{03} \cup U_{04}.\end{aligned}$$

For a supplied tuple $\boldsymbol{t} = (t_1, \dots, t_n)$, write $\mathcal{G}_n(\boldsymbol{t})$ if there is an ordered partition $\{1, \dots, n\} = J \sqcup K$ into two nonempty blocks such that, on putting $m = \sum_{j \in J} t_j$ and $h = \sum_{k \in K} t_k$, either

$$(m, h) \in V, \qquad \text{or} \qquad (m, h) \in U_{01} \quad \text{and} \quad \min(|J|, |K|) = 1.$$

This fixes the literal two-block "good" convention immediately preceding equation (12) in Li [3]. The larger assignment inventory in the verification appendix permits unused factors only to overcount possible boundary faces; it does not enlarge this numerical hypothesis's good sets.

Write $\beta = 1 - t_1 - t_2 - t_3$. The four integration domains are

$$\begin{aligned}
R_0 &= \Omega \setminus U_0,\\
R_1 &= \{\boldsymbol{t} \in \mathbb{R}^4 : (t_1,t_2) \in U_{03}, \quad a \le t_3 < \min(t_2, (1-s)/2), \quad \neg\mathcal{G}_3(t_1,t_2,t_3),\\
&\qquad\qquad a \le t_4 < \min(t_3, (1-s-t_3)/2), \quad \neg\mathcal{G}_4(t_1,t_2,t_3,t_4)\},\\
R_2 &= \{\boldsymbol{t} \in \mathbb{R}^4 : (t_1,t_2) \in U_{04}, \quad a \le t_3 < \min(t_2, (1-s)/2), \quad \neg\mathcal{G}_3(t_1,t_2,t_3),\\
&\qquad\qquad a \le t_4 < t_1/2, \quad \neg\mathcal{G}_4(\beta,t_2,t_3,t_4)\},\\
R_3 &= \{\boldsymbol{t} \in \mathbb{R}^3 : (t_1,t_2) \in R_0, \quad t_2 < t_3 < (1-s)/2, \quad \mathcal{G}_3(t_1,t_2,t_3)\}.
\end{aligned}$$

Let $\omega$ be Buchstab's function, defined by $\omega(u) = 1/u$ for $1 \le u \le 2$ and $(u\omega(u))' = \omega(u-1)$ for $u > 2$. Use the source's upper envelope

$$\omega_+(u) = \begin{cases} \omega(u), & 1 \le u < 4,\\ 5617/10000, & u \ge 4, \end{cases}$$

which is the function denoted $\omega_1$ in Li [3, Section 7]. Its upper-envelope property is the standard Buchstab bound used there; see also Harman [4]. All arguments of these functions below are at least one. With $\mathrm{d}\boldsymbol{t}_n = \mathrm{d}t_n \cdots \mathrm{d}t_1$, put

$$I_0 = \int_{R_0} \frac{\omega((1-t_1-t_2)/t_2)}{t_1 t_2^2}\,\mathrm{d}\boldsymbol{t}_2, \tag{3}$$

$$I_1 = \int_{R_1} \frac{\omega_+((1-t_1-t_2-t_3-t_4)/t_4)}{t_1 t_2 t_3 t_4^2}\,\mathrm{d}\boldsymbol{t}_4, \tag{4}$$

$$I_2 = \int_{R_2} \frac{\omega_+((t_1-t_4)/t_4)\,\omega_+((1-t_1-t_2-t_3)/t_3)}{t_2 t_3^2 t_4^2}\,\mathrm{d}\boldsymbol{t}_4, \tag{5}$$

$$I_3 = \int_{R_3} \frac{\omega((1-t_1-t_2-t_3)/t_3)}{t_1 t_2 t_3^2}\,\mathrm{d}\boldsymbol{t}_3. \tag{6}$$

**Hypothesis 1.3** (the four Li III numerical inequalities)**.** For exactly the preceding domains and good predicate,

$$I_1 < \frac{1}{50}, \qquad I_2 < \frac{3}{100}, \qquad I_0 < \frac{97}{100}, \qquad I_3 > \frac{1}{25}.$$

These are the four numerical bounds recorded in Li's equations (15), (23), (24), and (26), now isolated from the analytic assertions. They imply $1 - I_1 - I_2 - I_0 + I_3 > 1/50$. The present paper gives exact certificates for the finite algebraic and geometric steps. It also reports an exploratory numerical recomputation of $I_0$, but no rigorous enclosure of that integral. $I_1$, $I_2$, and $I_3$ have not been independently recomputed here. Thus none of the four inequalities in Hypothesis 1.3 is replaced by an independent quadrature certificate in this paper.

## 2 Harmonic branching and compactness

The analytic estimates will control the total weight of defective integers at a given scale. They need not identify a good continuation below each individual prime. The following two results isolate the global argument that makes such information sufficient.

**Definition 2.1** (Branching system)**.** A branching system consists of a finite nonempty root set $V_1 \subset \mathbb{Z}_{\ge 2}$, good sets $G_n \subset \mathbb{Z}_{\ge 2}$, and finite child sets $\mathrm{Ch}_n(x) \subset \mathbb{Z}_{\ge 2}$. At each fixed level, child sets belonging to distinct parents are disjoint. Put

$$V_{n+1} = \bigcup_{x \in V_n \cap G_n} \mathrm{Ch}_n(x), \qquad w(S) = \sum_{x \in S} \frac{1}{x}$$

for finite sets $S$. An infinite branch is a sequence $(x_n)$ with $x_n \in V_n$ and $x_{n+1} \in \mathrm{Ch}_n(x_n)$ for every $n$.

**Theorem 2.2** (Mass–compactness principle)**.** *Suppose $\lambda_n > 0$ and $\beta_n \geq 0$ satisfy*

$$w(\mathrm{Ch}_n(x)) \geq \frac{\lambda_n}{x} \quad (x \in V_n \cap G_n), \qquad w(V_n \setminus G_n) \leq \beta_n.$$

*Define*

$$M_n = w(V_1) \prod_{k<n} \frac{\lambda_k}{2}.$$

*If $\beta_n \leq M_n/2$ for every $n$, then $w(V_n) \geq M_n > 0$ for every $n$, and the system has an infinite branch. Every node on such a branch is good at its own level.*

*Proof.* The assertion $w(V_1) \geq M_1$ is an equality. If $w(V_n) \geq M_n$, then

$$w(V_n \cap G_n) \geq w(V_n) - \beta_n \geq w(V_n) - M_n/2 \geq w(V_n)/2.$$

Disjointness of the child sets therefore gives

$$w(V_{n+1}) \geq \lambda_n w(V_n \cap G_n) \geq \frac{\lambda_n}{2} w(V_n) \geq M_{n+1}.$$

Thus every level is nonempty. Each level is finite, each nonroot node has a unique parent, and the root set is finite. After adjoining a common root, König's lemma gives an infinite path. A node can have a child in this forest only if it belongs to its good set. □

**Corollary 2.3** (Binary subdivision fixed by finite prefixes)**.** *Under the two local and defect bounds in Theorem 2.2, put*

$$s_n = \frac{w(V_1)}{3} \prod_{k<n} \frac{\lambda_k}{6}.$$

*Suppose, for every $n$,*

$$\beta_n \leq s_n/2, \qquad \max_{x \in V_n} \frac{1}{x} \leq s_n, \tag{B}$$

*where the second condition is vacuous for an empty set. Then the system has at least $2^{\aleph_0}$ distinct infinite branches.*

*Proof.* A finite collection of weight at least $3s$, with each atom of weight at most $s$, splits into two disjoint parts of weight at least $s$. Indeed, in increasing integer order, stop at the first partial sum at least $s$. Its weight is at most $2s$, so the remaining weight is at least $s$.

Fix these splits once for each finite binary word; they are not chosen separately for different infinite addresses. For a word $\sigma$ of length $n-1$, construct $U_\sigma \subset V_n$ with $w(U_\sigma) \geq 3s_n$, starting with $U_\varnothing = V_1$. Split $U_\sigma$ into $S_{\sigma,0}, S_{\sigma,1}$, each of weight at least $s_n$. For $b \in \{0,1\}$ define

$$U_{\sigma b} = \bigcup_{x \in S_{\sigma,b} \cap G_n} \mathrm{Ch}_n(x).$$

The defective weight in any selected subset is bounded by the defective weight of the entire level. Hence

$$w(U_{\sigma b}) \geq \lambda_n (w(S_{\sigma,b}) - \beta_n) \geq \lambda_n s_n/2 = 3s_{n+1}.$$

The atom bound in (B) permits the next split. Induction defines all finite-prefix families.

For an infinite address $\omega = (\omega_1, \omega_2, \ldots)$, the selected level-$n$ set is $S_{\omega|_{n-1}, \omega_n}$. These sets form a locally finite forest with finitely many roots and a nonempty set at every depth. König's lemma supplies an infinite branch in that forest. If two addresses first differ at position $n$, their level-$n$ nodes belong to the two disjoint parts of the same $U_\sigma$. Their branches are therefore distinct. For definiteness, a branch for each address can be selected by taking, at each step, the least node that has descendants at arbitrarily large depths. This also makes the simultaneous choice explicit. □

*Remark* 2.4. Neither result assumes that each good parent has a good child. The defect bound is global, and remains valid after any of the finite-prefix selections. Concentrating defective nodes beneath one parent does not invalidate the comparison.

## 3 From almost-all short intervals to full prime cells

We formulate the prime input using centered intervals solely to accommodate both orientations of the analytic estimates. A lower bound in either $(u - X^\theta, u]$ or $(u, u + X^\theta]$ implies the centered lower bound below by inclusion. The same holds for a multiplicative window of length at most $X^\theta$, provided its lower bound is a fixed positive multiple of $X^\theta/\log X$.

**Hypothesis 3.1** (AA($\theta$))**.** Fix $0 < \theta < 1$. For every fixed $D > 0$, there are $\kappa_D > 0$ and $X_D$ such that, for all real $X \geq X_D$,

$$\operatorname{meas} E_{\theta,D}(X) \leq X \exp[-D(\log\log X)^2], \tag{AA}$$

where

$$E_{\theta,D}(X) = \left\{ u \in [X, 2X] : \pi(u + X^\theta) - \pi(u - X^\theta) < \kappa_D \frac{X^\theta}{\log X} \right\}.$$

The lower-bound constant is allowed to depend on the fixed accuracy $D$. This causes no difficulty: the construction below chooses $D$ once, before choosing the starting scale. An implicit fixed multiplicative constant in (AA) is harmless, since it is absorbed by first using a larger fixed accuracy.

**Lemma 3.2** (A poor cell uses a positive proportion of its length)**.** *Let $h = X^\theta$ and let $E \subset [X, 2X]$ be an exceptional set outside which*

$$\pi(u + h) - \pi(u - h) \geq \kappa h/\log X.$$

*If $I = [a, b) \subset [X, 2X]$, $L = b - a \geq 4h$, and*

$$\#(\mathcal{P} \cap I) < \frac{\kappa L}{16 \log X},$$

*then* $\operatorname{meas}(E \cap I) \geq L/4$.

*Proof.* For $u \in B_I = [a + h, b - h)$, the centered window is contained in $I$, apart from endpoints of measure zero. Each prime of $I$ is counted for at most $2h$ units of $u$-measure. Thus

$$\int_{B_I} (\pi(u + h) - \pi(u - h))\, du \leq 2h\, \#(\mathcal{P} \cap I) < \frac{\kappa h L}{8 \log X}.$$

The nonexceptional part of $B_I$ has measure at most $L/8$. Since $\operatorname{meas} B_I = L - 2h \geq L/2$, the exceptional part has measure at least $3L/8$, which is more than required. □

For $c > 1$ and an integer $m \geq 2$ define

$$J_c(m) = [m^c, (m + 1)^c - 1), \qquad \mathcal{P}_c(m) = \mathcal{P} \cap J_c(m). \tag{J}$$

For a fixed $c$, these cells are pairwise disjoint. The mean value theorem gives

$$c\, m^{c-1} - 1 \leq |J_c(m)| \leq c\, (m + 1)^{c-1}. \tag{L}$$

**Proposition 3.3** (Uniform bad-integer estimate)**.** *Assume* AA($\theta$)*, and fix*

$$\frac{1}{1 - \theta} < c_- \leq c_+ < \infty.$$

*For every fixed $D > 0$, there are $\delta_D \in (0, 1]$ and $Y_D$ such that, for every real $Y \geq Y_D$ and every $c \in [c_-, c_+]$,*

$$\# \left\{ m \in [Y, 2Y) \cap \mathbb{Z} : \#\mathcal{P}_c(m) < \delta_D \frac{|J_c(m)|}{\log((m + 1)^c)} \right\} \leq Y \exp[-D(\log\log Y)^2]. \tag{N}$$

*Proof.* Use Hypothesis 3.1 with a fixed accuracy $A > D$. All cells under consideration lie in $[Y^c, (2Y+1)^c]$, which is covered by $O_{c_+}(1)$ successive dyadic intervals $[X, 2X)$ beginning at $X = Y^c$. At most $O_{c_+}(1)$ cells cross boundaries of this cover, because the cells are disjoint. Set those cells aside.

For every remaining cell $I = J_c(m) \subset [X, 2X]$,

$$|I| \asymp_{c_-,c_+} Y^{c-1}, \qquad X \asymp_{c_+} Y^c, \qquad \frac{|I|}{X^\theta} \gg Y^{\,c_-(1-\theta)-1} \longrightarrow \infty.$$

All comparisons are uniform in $c \in [c_-, c_+]$. Choose $\delta_D = \min(1, \kappa_A/16)$. Since $\log((m+1)^c) \geq \log X$, a cell counted on the left of (N) satisfies the poor-cell hypothesis of Lemma 3.2. It accounts for at least $|I|/4$ of exceptional endpoint measure in its own dyadic range. Disjointness therefore bounds the number of such cells by

$$O_{c_-,c_+}\left(Y \exp[-A(\log\log(Y^c))^2]\right).$$

Now $\log\log(Y^c) = \log\log Y + \log c$, with $c$ in a fixed compact interval above 1. Because $A > D$, the fixed constants, the finite cover, and the $O_{c_+}(1)$ discarded boundary cells are absorbed in the right side of (N) for large $Y$. □

*Remark* 3.4. The averaging step charges a poor cell its full length $\asymp Y^{c-1}$. Charging only one interval of length $X^\theta$ would give a weaker count and would not prove (N). There is no conversion of a fixed-logarithm theorem with a moving logarithmic exponent in this argument.

## 4 The harmonic transfer theorem

For a sequence $\mathbf{c} = (c_k)_{k\geq 1}$ with $c_k > 1$, put

$$C_n = \prod_{k=1}^{n} c_k, \qquad \mathcal{W}(\mathbf{c}) = \{A > 1 : \lfloor A^{C_n} \rfloor \in \mathcal{P} \text{ for every } n \geq 1\}.$$

The constant-sequence case is denoted by $\mathcal{W}(c)$.

**Lemma 4.1** (The strict upper endpoint)**.** *Suppose $p_n \geq 2$ and $p_{n+1} \in J_{c_{n+1}}(p_n)$ for every $n \geq 1$. Then*

$$A = \lim_{n\to\infty} p_n^{1/C_n}$$

*exists, is greater than 1, and satisfies $\lfloor A^{C_n} \rfloor = p_n$ for every $n$. Distinct sequences $(p_n)$ give distinct values of $A$.*

*Proof.* Put $a_n = p_n^{1/C_n}$ and $b_n = (p_n+1)^{1/C_n}$. The child inequalities give

$$a_{n+1} \geq a_n, \qquad b_{n+1} < b_n.$$

The first sequence is increasing and bounded above by $b_1$, so its limit $A$ exists. For each fixed $n$,

$$a_n \leq A \leq b_{n+1} < b_n. \tag{E}$$

Consequently $p_n \leq A^{C_n} < p_n + 1$, including the required strict upper floor inequality. Also $A \geq p_1^{1/c_1} > 1$. An $A$ uniquely determines every integer $\lfloor A^{C_n} \rfloor$, which proves distinctness. □

**Proposition 4.2** (Nullity and category need no prime input)**.** *More generally, for any sequence of positive exponents $C_n \to \infty$, the set*

$$\{A > 1 : \lfloor A^{C_n} \rfloor \in \mathcal{P} \text{ for every } n\}$$

*has Lebesgue measure zero and is nowhere dense. Its closure also has measure zero.*

*Proof.* Fix $1 < a < b$. The set satisfying the $n$th prime condition in $[a, b]$ is a finite union of intersections with intervals $[p^{1/C_n}, (p+1)^{1/C_n})$, where $a^{C_n} - 1 \le p \le b^{C_n}$. For large $n$, each such interval has length at most $b/(C_n p)$ by the mean value theorem. Mertens' theorem gives

$$\frac{b}{C_n} \sum_{a^{C_n}-1 \le p \le b^{C_n}} \frac{1}{p} = \frac{b}{C_n} \left( \log \frac{\log b}{\log a} + o(1) \right) \longrightarrow 0.$$

Closing a finite union of these intervals adds only finitely many endpoints. The closure of the set satisfying every prime condition in $[a, b]$ is contained in each of these closed unions, and is therefore null. To control the closure of the full set on a given compact subinterval of $(1, \infty)$, apply this argument on a slightly larger compact interval. A countable cover, together with the possible boundary point 1, proves that the full closure is null. A closed null set has empty interior. □

**Theorem 4.3** (Transfer from one quantitative bad-node estimate)**.** *Let $1 < c_- \le c_+ < \infty$. Suppose there are constants $\delta \in (0, 1]$, $K > 0$, $Y_0$ and*

$$D > \frac{1}{2 \log c_-} \tag{T}$$

*such that, for every $Y \ge Y_0$ and every $c \in [c_-, c_+]$,*

$$\# \left\{ m \in [Y, 2Y) \cap \mathbb{Z} : \#\mathcal{P}_c(m) < \delta \frac{|J_c(m)|}{\log((m+1)^c)} \right\} \le KY \exp[-D(\log\log Y)^2]. \tag{S}$$

*Then, for every sequence $\mathbf{c}$ taking values in $[c_-, c_+]$, $\mathcal{W}(\mathbf{c})$ has cardinality $2^{\aleph_0}$, is unbounded, has Lebesgue measure zero, and is nowhere dense.*

*Proof.* Fix the constants in the theorem and then choose a large $P$. No parameter below depends on the eventual depth of the tree. Let

$$V_1 = \mathcal{P} \cap [P, 2P], \qquad \mathrm{Ch}_n(m) = \mathcal{P}_{c_{n+1}}(m).$$

Call $m$ good at level $n$ when it satisfies the inequality opposite to that defining the exceptional set in (S), with $c = c_{n+1}$. Write

$$R_1 = 1, \qquad R_n = c_2 \cdots c_n = C_n / c_1.$$

**Ranges and branching weights.** Every $p \in V_n$ satisfies

$$P^{R_n} \le p, \qquad p + 1 \le (2P+1)^{R_n}. \tag{R}$$

Indeed, the lower inequality propagates through $q \ge p^{c_{n+1}}$, and the upper one propagates through $q + 1 < (p+1)^{c_{n+1}}$. These are direct inductions from $V_1$. For a good parent $m$, (L) and the definition of goodness give, uniformly for $c = c_{n+1} \in [c_-, c_+]$,

$$\begin{aligned} w(\mathrm{Ch}_n(m)) &\ge \frac{\delta\,(c\,m^{c-1} - 1)}{c \log(m+1)(m+1)^c} \\ &\ge \frac{\delta}{4m \log(m+1)} \ge \frac{\lambda_n}{m}, \qquad \lambda_n = \frac{\delta}{4R_n \log(4P)}. \end{aligned} \tag{W}$$

The second inequality holds for a single sufficiently large $P$, uniformly on the fixed compact range of $c$. The prime number theorem also gives

$$w(V_1) \ge \frac{1}{4 \log P}. \tag{P}$$

**Defect weights.** The range in (R) is covered by at most $3R_n$ successive dyadic intervals whose lower endpoints are at least $P^{R_n}$, once $P \geq 2$. Divide (S) by the lower endpoint of each interval and sum. With $\ell = \log P$ and $L = \log \ell$, this gives the uniform upper bound

$$w(V_n \setminus G_n) \leq 3KR_n \exp[-D(L + \log R_n)^2] =: \beta_n. \tag{D}$$

All analytic hypotheses apply at all levels as soon as $P \geq Y_0$; a new scale threshold is not chosen for each $n$.

**One starting scale for every depth.** Use the reserves $s_n$ from Corollary 2.3. Put $a = \log c_- > 0$, $r_n = \log R_n$, and $j = n - 1$. Since consecutive increments of $r_n$ are at least $a$,

$$j \leq r_n/a, \qquad r_n - r_k \geq (n-k)a \quad (k < n).$$

In particular,

$$\sum_{k<n} r_k \leq jr_n - \frac{aj(j+1)}{2} \leq \frac{r_n^2}{2a}. \tag{Q}$$

The last inequality follows by completing the square and dropping the nonpositive term $-aj/2$. Equations (W) and (P) therefore imply

$$-\log s_n \leq L + O(1) + j(L + O(1)) + \sum_{k<n} r_k \leq \frac{r_n^2}{2a} + \frac{r_n L}{a} + O(r_n + L + 1). \tag{M}$$

All implied constants here are fixed and independent of $n$, $P$ and the particular sequence in $[c_-, c_+]$. Combining (D) and (M) yields

$$\begin{aligned} \log(\beta_n/s_n) &\leq -D(L + r_n)^2 + \frac{r_n^2}{2a} + \frac{r_n L}{a} + O(r_n + L + 1) \\ &= -\left(D - \frac{1}{2a}\right)(L + r_n)^2 - \frac{L^2}{2a} + O(r_n + L + 1). \end{aligned} \tag{U}$$

By (T), the coefficient of the first quadratic term is strictly negative. Since $L + r_n \geq L$, choosing $L$ sufficiently large makes this upper bound at most $-\log 2$ simultaneously for every $r_n \geq 0$. Thus $\beta_n \leq s_n/2$ at every level.

The atom bound is also uniform. By (R),

$$\max_{p \in V_n} \frac{1}{p} \leq \exp(-\ell e^{r_n}).$$

For $r \geq 0$, $e^r \geq 1 + r + r^2/2$. Because $\ell = e^L$, a single sufficiently large $L$ makes the coefficients of $\ell(1 + r + r^2/2)$ dominate those of

$$\frac{r^2}{2a} + \frac{rL}{a} + O(r + L + 1)$$

for every $r \geq 0$. Consequently the last displayed atom bound is at most $s_n$ for all $n$. Both conditions in (B) now hold with the same $P$.

**Branches, representing constants, and size.** Corollary 2.3 supplies $2^{\aleph_0}$ distinct prime branches. By Lemma 4.1, each gives a distinct member of $\mathcal{W}(\mathbf{c})$. Since this is a subset of $\mathbb{R}$, its cardinality is exactly $2^{\aleph_0}$. Every such constant satisfies $A \geq P^{1/c_1}$. The construction works for all sufficiently large $P$, so $\mathcal{W}(\mathbf{c})$ is unbounded. Finally $C_n \geq c_-^n \to \infty$, and Proposition 4.2 supplies nullity and nowhere density. □

*Remark* 4.4 (Comparison with the earlier sufficient threshold). The earlier comparison bounded $R_n$ above and below separately and imposed

$$D > \frac{\log c_+}{2(\log c_-)^2}.$$

That condition remains sufficient: $\log c_+ \geq \log c_-$ implies that it is at least as strong as (T). It also forces the mixed coefficient $2D \log c_- - 1$ to be positive. The sharper estimate (Q) compares the accumulated loss with the actual current scale and avoids the artificial $c_+$ loss. This refines the required exceptional accuracy, without changing the prime-interval exponent.

**Corollary 4.5** (Transfer of a quantitative almost-all prime input)**.** *Assume* $\mathsf{AA}(\theta)$*. If* $\mathbf{c}$ *satisfies*

$$\frac{1}{1-\theta} < \inf_k c_k \leq \sup_k c_k < \infty,$$

*then* $\mathcal{W}(\mathbf{c})$ *has cardinality* $2^{\aleph_0}$*, is unbounded, is Lebesgue-null, and is nowhere dense.*

*Proof.* Set $c_- = \inf c_k$ and $c_+ = \sup c_k$. Choose one fixed $D > 1/(2 \log c_-)$. Proposition 3.3 supplies (S) for this $D$, with a fixed $\delta_D > 0$. Apply Theorem 4.3. □

*Remark* 4.6 (Quantifiers and endpoints). The order is: fix the exponent range, choose the exceptional accuracy $D$, obtain the fixed prime lower-bound constant, and then choose one starting scale $P$ for the whole infinite tree. A bound $O_B(X \log^{-B} X)$ separately for each fixed $B$ does not by itself justify this order, or permit replacing $B$ by a function of $X$ or the depth. The strict condition $\theta < 1 - 1/c_-$ is used in the full-cell averaging step. Thus an input at exactly $\theta$ gives $c > 1/(1-\theta)$. An improvement to $\theta < 1/24$ includes $c = 24/23$ because the needed inequality is then strict. The stronger Vinogradov–Korobov exceptions in Section 5 also meet the quantitative node hypothesis.

## 5 Primes in almost all short intervals with Vinogradov–Korobov savings

Let $N(\sigma, T)$ denote the number of zeros $\rho = \beta + i\gamma$ of $\zeta(s)$ with $\beta \geq \sigma$ and $|\gamma| \leq T$. Write $\mathcal{L}(X) = (\log X)^{1/3} (\log \log X)^{-1/3}$.

**Hypothesis 5.1** (Z($\theta_1$))**.** Let $0 < \theta_1 < 1$. There are $\sigma_1 \in [1/2, 1)$, $\delta_0 > 0$ and $B \geq 0$ such that, with $a_* = 2/(1-\theta_1) - \delta_0$,

(a) for every $\varepsilon > 0$: $N(\sigma, T) \ll_\varepsilon T^{a_*(1-\sigma)+\varepsilon}$ uniformly for $1/2 \leq \sigma \leq \sigma_1$, $T \geq 2$;

(b) $N(\sigma, T) \ll T^{a_*(1-\sigma)} (\log T)^B$ uniformly for $\sigma_1 \leq \sigma \leq 1$, $T \geq 2$.

We use the following classical facts.

(F1) Explicit formula [5, Ch. 17]: for $x \geq 2$ and $T \geq 2$, $\psi(x) = x - \sum_{|\gamma|<T} x^\rho/\rho + O\big(x(\log xT)^2/T + \log x\big)$.

(F2) Zero counting [5, Ch. 15]: $N(t+1) - N(t) \ll \log(|t|+2)$ for the number $N(t)$ of zeros with $0 < \gamma \leq t$.

(F3) Vinogradov–Korobov [6, Thm. 6.1], [7]: there is $c_0 > 0$ such that $\zeta(\sigma + it) \neq 0$ for $\sigma \geq 1 - c_0 (\log |t|)^{-2/3} (\log \log |t|)^{-1/3}$, $|t| \geq 3$.

(F4) Ingham [8], [6, Ch. 11]: $N(\sigma, T) \ll T^{3(1-\sigma)/(2-\sigma)} (\log T)^5$ for $1/2 \leq \sigma \leq 1$.

(F5) Huxley [9], [6, Ch. 11]: $N(\sigma, T) \ll T^{3(1-\sigma)/(3\sigma-1)} (\log T)^{44}$ for $3/4 \leq \sigma \leq 1$.

(F6) Guth–Maynard [10]: for each fixed $\sigma \in [7/10, 8/10]$ and every $\varepsilon > 0$, $N(\sigma, T) \ll_{\sigma,\varepsilon} T^{15(1-\sigma)/(3+5\sigma)+\varepsilon}$.

**Lemma 5.2** (verification of $\mathrm{Z}(\theta_1)$)**.** (i) $\mathrm{Z}(\theta_1)$ *holds for every* $\theta_1 > 1/6$*, using* (F4) *and* (F5) *only.*

(ii) $\mathrm{Z}(\theta_1)$ *holds for every* $\theta_1 > 2/15$*, using* (F4), (F5) *and* (F6)*.*

(iii) *Under the density hypothesis,* $\mathrm{Z}(\theta_1)$ *holds for every* $\theta_1 > 0$*.*

*Proof.* (i) On $[1/2, 3/4]$ the exponent $3/(2-\sigma)$ of (F4) is at most $12/5$, and on $[3/4, 1]$ the exponent $3/(3\sigma-1)$ of (F5) is at most $12/5$. Hence $N(\sigma, T) \ll T^{12(1-\sigma)/5}(\log T)^{44}$ on $[1/2, 1]$. If $\theta_1 > 1/6$ then $12/5 < 2/(1-\theta_1)$; take $\sigma_1 = 1/2$ and $\delta_0 = 2/(1-\theta_1) - 12/5$ in (b).

(ii) Let $\theta_1 > 2/15$, so that $30/13 < 2/(1-\theta_1)$. On $[1/2, 7/10]$, (F4) gives exponent $3/(2-\sigma) \le 30/13$ with a logarithmic factor; on $[7/10, 4/5]$, (F6) gives $15/(3+5\sigma) \le 30/13$ with a factor $T^{\varepsilon}$; on $[4/5, 1]$, (F5) gives $3/(3\sigma-1) \le 15/7 < 30/13$ with a logarithmic factor. Take $\sigma_1 = 4/5$ and $\delta_0 = \frac{1}{2}(2/(1-\theta_1) - 30/13)$, so that $a_* = 30/13 + \delta_0$. Then (b) holds on $[4/5, 1]$ directly from (F5) with $B = 44$, because $15/7 < a_*$; on $[1/2, 7/10]$ the logarithmic factor in (F4) is $\ll_{\varepsilon} T^{\varepsilon}$, which gives (a) there. For (a) on $[7/10, 4/5]$, where (F6) is a pointwise statement, use that $N(\sigma, T)$ is nonincreasing in $\sigma$: for a mesh $7/10 = s_0 < s_1 < \cdots < s_r = 4/5$ with $s_{j+1} - s_j \le \delta_0/20$ and $\sigma \in [s_j, s_{j+1}]$, $N(\sigma, T) \le N(s_j, T) \ll T^{\frac{30}{13}(1-s_j)+\varepsilon} \le T^{\frac{30}{13}(1-\sigma)+\frac{30}{13}\cdot\frac{\delta_0}{20}+\varepsilon}$, and the extra exponent is absorbed into $\delta_0(1-\sigma)$ because $1-\sigma \ge 1/5$ and $\frac{30}{13}\cdot\frac{1}{20} < \frac{1}{5}$.

(iii) Let $\theta_1 > 0$. The density hypothesis gives (a) with exponent $2 < 2/(1-\theta_1)$ on $[1/2, 1]$. For (b) choose $\sigma_1 \in [3/4, 1)$ with $3/(3\sigma_1 - 1) < 2/(1-\theta_1)$, which is possible because $3/(3\sigma-1) \to 3/2$ as $\sigma \to 1$, and use (F5) on $[\sigma_1, 1]$. Take $\delta_0$ smaller than both gaps. If the density hypothesis is only available pointwise in $\sigma$, the mesh argument of (ii) gives the required uniformity. □

**Proposition 5.3** (mean square with VK savings; classical, cf. [11, 12, 13, 14])**.** *Assume* $\mathrm{Z}(\theta_1)$ *and let* $\theta_1 \le \theta_2 < 1$*. There are* $\kappa > 0$ *and* $X_0$ *such that for* $X \ge X_0$ *and* $X^{\theta_1} \le h \le X^{\theta_2}$*,*

$$\int_X^{2X} \bigl|\psi(u+h) - \psi(u) - h\bigr|^2 \, du \le h^2 X \exp(-\kappa\, \mathcal{L}(X)).$$

*Proof.* Let $T = X^{1-\theta_1}\exp(2\sqrt{\log X})$, replacing it if necessary by a height in $[T, T+1]$ away from zero ordinates; then $2 \le T \le X$ for large $X$. By (F1), for $u \in [X, 2X]$,

$$\psi(u+h) - \psi(u) - h = -S(u) + O\Bigl(\frac{X\log^2 X}{T} + \log X\Bigr),$$
$$S(u) = \sum_{|\gamma|<T} \frac{(u+h)^{\rho} - u^{\rho}}{\rho} = \int_u^{u+h} F(v)\, dv,$$

with $F(v) = \sum_{|\gamma|<T} v^{\rho-1}$. The error term is at most $h\exp(-\sqrt{\log X})$ for large $X$, since $X\log^2 X/T = X^{\theta_1}\log^2 X\, e^{-2\sqrt{\log X}}$ and $h \ge X^{\theta_1}$.

By Cauchy–Schwarz, $|S(u)|^2 \le h\int_u^{u+h} |F(v)|^2 dv$, and integrating over $u \in [X, 2X]$ (each $v$ lies in at most $h$ units of $u$-measure),

$$\int_X^{2X} |S(u)|^2 du \le h^2 \int_X^{3X} |F(v)|^2 dv = h^2 \sum_{\rho,\rho'} \int_X^{3X} v^{\rho+\overline{\rho'}-2} dv.$$

For $\rho = \beta + i\gamma$ and $\rho' = \beta' + i\gamma'$ the inner integral is at most $2X^{\beta+\beta'-1}$ in absolute value, and also at most $6X^{\beta+\beta'-1}/|\gamma-\gamma'|$ by integrating the oscillating factor. Hence it is $\le 12X^{\beta+\beta'-1}/(1+|\gamma-\gamma'|)$. With $X^{\beta+\beta'} \le (X^{2\beta} + X^{2\beta'})/2$ and symmetry,

$$\int_X^{3X} |F|^2 \le 12 \sum_{|\gamma|<T} X^{2\beta-1} \sum_{|\gamma'|<T} \frac{1}{1+|\gamma-\gamma'|} \ll (\log T)^2 \sum_{|\gamma|<T} X^{2\beta-1},$$

by (F2). Consequently

$$\int_X^{2X} |\psi(u+h)-\psi(u)-h|^2 du \ll h^2 X\Big((\log X)^2\Sigma + e^{-2\sqrt{\log X}}\Big), \qquad \Sigma = \sum_{|\gamma|<T} X^{2\beta-2}.$$

Zeros with $\beta < 1/2$ contribute at most $N(T)X^{-1} \ll T\log T/X \le X^{-\theta_1/2}$ to $\Sigma$. For $\beta \ge 1/2$, partial summation gives

$$\sum_{\beta\ge 1/2} X^{2\beta-2} = X^{-1}N(\tfrac12,T) + 2\log X \int_{1/2}^{1} X^{2\sigma-2}N(\sigma,T)\,d\sigma.$$

The endpoint term $X^{-1}N(1/2,T) \ll T\log T/X$ has the same fixed power saving as the zeros with $\beta < 1/2$. By (F3), after decreasing $c_0$ to cover the finitely many bounded-height zeros, $N(\sigma,T) = 0$ for $\sigma > 1-\eta$ with $\eta = c_0(\log T)^{-2/3}(\log\log T)^{-1/3}$. Write $a_*(1-\theta_1) = 2-\delta_1$ with $\delta_1 = \delta_0(1-\theta_1) > 0$. Then $T^{a_*(1-\sigma)}X^{2\sigma-2} = \exp(-(1-\sigma)[\delta_1\log X - 2a_*\sqrt{\log X}]) \le \exp(-(1-\sigma)\frac{\delta_1}{2}\log X)$ for large $X$. On $[\sigma_1, 1-\eta]$ hypothesis (b) bounds the integrand by $(\log X)^B\exp(-\frac{\delta_1}{2}\eta\log X)$. On $[1/2,\sigma_1]$ hypothesis (a), with $\varepsilon = (1-\sigma_1)\delta_1/4$, bounds it by $X^{-(1-\sigma_1)\delta_1/4}$. Since $\log T \asymp \log X$, we have $\eta\log X \gg \mathcal{L}(X)$, and therefore $\Sigma \ll (\log X)^{B+1}\exp(-c'\mathcal{L}(X))$ for some $c' > 0$. The proposition follows with any $\kappa < c'$. □

*Proof of Theorem 1.2.* For the applicable density threshold $\theta_0 \in \{1/6, 2/15, 0\}$, choose fixed exponents $\theta_0 < \theta_1 < \vartheta < 1 - 1/c_-$ and then $\vartheta < \theta_2 < 1$. The preceding lemma verifies $\mathrm{Z}(\theta_1)$, with the density hypothesis assumed only in the third case. Apply the mean-square proposition with $h = X^{\vartheta}$. Chebyshev's inequality gives a prime-weighted lower bound outside measure $O(Xe^{-a\mathcal{L}(X)})$ for some $a > 0$.

Prime powers of exponent at least two contribute uniformly

$$O(hX^{-1/2}(\log X)^2 + (\log X)^2) = o(h)$$

to the difference of $\psi$ on an interval of length $h$ at height comparable to $X$: count the possible $k$th roots in each interval and sum over $2 \le k \ll \log X$. Thus the number of primes is at least a fixed positive multiple of $h/\log X$. Translation and a bounded number of comparable dyadic ranges give the same assertion for backward windows $[x - X^{\vartheta}, x]$. Since $e^{-a\mathcal{L}(X)}$ is smaller than $\exp[-D(\log\log X)^2]$ for every fixed $D$, the quantitative short-interval hypothesis in Section 4 applies. Its bounded-variable-exponent conclusion proves all three assertions. □

## 6 Why fixed logarithmic estimates cannot simply be substituted

For each separately fixed $B$, the stated almost-all sieve theorem supplies exceptional measure $O_B(X(\log X)^{-B})$. The harmonic construction needs a uniform estimate of the form

$$X\exp(-D(\log\log X)^2).$$

The first assertion alone does not imply the second. For example,

$$r(X) = \exp(-(\log\log X)^{3/2})$$

is $O_B((\log X)^{-B})$ for every fixed $B$, but exceeds $\exp(-D(\log\log X)^2)$ eventually for every fixed $D > 0$.

This distinction has a concrete tree interpretation. At level $n$, $\log Y_n$ is comparable to $c^n$, and repeatedly retaining a prime proportion of size $1/\log Y_n$ loses

$$\prod_{j<n}\frac{1}{\log Y_j} = \exp\left(-\frac{\log c}{2}n^2 + O(n)\right).$$

A fixed logarithmic exceptional rate gives only $\exp(-Bn\log c+O(1))$, which eventually exceeds this retained mass. The analytic proof must therefore be made quantitative before it is combined with the tree argument.

For $D>0$ define, for sufficiently large $X$,

$$\mathcal{Q}_D(X)=\exp(D(\log\log X)^2),\qquad \mathcal{L}(X)=\frac{(\log X)^{1/3}}{(\log\log X)^{1/3}}.$$

For any fixed $D,K,d,\nu>0$,

$$(\log X)^d=\mathcal{Q}_D(X)^{o(1)},\quad \mathcal{Q}_D(X)^K=X^{o(1)},\quad \mathcal{Q}_D(X)^K e^{-\nu\mathcal{L}(X)}=o(\mathcal{Q}_D(X)^{-d}). \tag{7}$$

The order of choices below is essential: fix the interval exponent, then fix all sieve and moment parameters, then choose the finite accuracy powers, and finally let $X$ increase.

# 7 A quantitative prime-polynomial estimate

**Lemma 7.1** (uniform prime-polynomial bound)**.** *Fix $\delta>0$, $M_0>0$, and a bounded real interval of values of $\sigma$. There is $a>0$ such that, whenever $X$ is sufficiently large, $X^\delta\le u\le v\le 2u\ll X$, and $|t|\le X^{M_0}$,*

$$\sum_{u<n\le v}\frac{\Lambda(n)}{n^{\sigma+it}}=\int_u^v y^{-\sigma-it}\,dy+O_{\delta,M_0,\sigma}\left(u^{1-\sigma}e^{-a\mathcal{L}(X)}\right).$$

*The constants are uniform when $\sigma$ stays in the indicated compact interval. Consequently*

$$\left|\sum_{u<n\le v}\frac{\Lambda(n)}{n^{\sigma+it}}\right|\ll u^{1-\sigma}\left(\frac{1}{1+|t|}+e^{-a\mathcal{L}(X)}\right).$$

*The analogous upper bound for primes with coefficient 1 holds, with at most a fixed logarithmic factor.*

*Proof.* Let $s=\sigma+it$. Apply truncated Perron's formula to $-\zeta'(s+w)/\zeta(s+w)$ with difference kernel

$$\frac{v^w-u^w}{w}.$$

Take the initial line with $\Re(s+w)>1$, and choose a truncation height $X^M$ with fixed $M$ sufficiently large relative to $M_0$ and the allowed range of $\sigma$. The endpoint convention for Perron's formula changes the normalized sum by $O(u^{-\sigma}\log X)$, which is absorbed by the asserted error.

The Vinogradov–Korobov zero-free region [7, 6] permits a shift to $\Re(s+w)=1-\eta$, where

$$\eta=b_0(\log X)^{-2/3}(\log\log X)^{-1/3}$$

and $b_0>0$ is sufficiently small. Decreasing $b_0$ handles bounded heights as well. In this narrower zero-free strip, $-\zeta'/\zeta(z)-(z-1)^{-1}$ has a fixed power of $\log X$ as an upper bound. The only pole crossed is at $w=1-s$, and its residue is

$$\frac{v^{1-s}-u^{1-s}}{1-s}=\int_u^v y^{-s}\,dy.$$

The difference kernel is regular at $w=0$, so it creates no additional residue there. The shifted vertical integral is

$$\ll u^{1-\sigma-\eta}(\log X)^{O(1)}\ll u^{1-\sigma}e^{-a\mathcal{L}(X)},$$

because $\log u\ge\delta\log X$. The horizontal integrals and Perron truncation error are smaller on increasing $M$. This proves the expansion.

For the main integral, writing $y=ue^z$ shows that its size is $O(u^{1-\sigma}/(1+|t|))$, uniformly for $0\le\log(v/u)\le\log 2$. Removing prime powers costs at most a fixed logarithmic factor times $u^{1/2-\sigma}+u^{-\sigma}$, which is negligible here. Partial summation removes the remaining $\log p$ weight. □

**Corollary 7.2.** *For every fixed $D, K > 0$, the prime polynomials in Lemma 7.1 satisfy*

$$\left| \sum_{u<n\le v} \Lambda(n) n^{-\sigma-it} \right| \ll u^{1-\sigma} \mathcal{Q}_D(X)^{-K}$$

*throughout $\mathcal{Q}_D(X)^{K+1} \le |t| \le X^{M_0}$, for sufficiently large $X$. The low-frequency main integral is retained rather than discarded.*

**Lemma 7.3** (fixed products of large primes)**.** *The same conclusion, with a possible fixed logarithmic loss, holds for Dirichlet polynomials formed from a fixed number of prime variables $p_j \ge X^{\delta}$, with fixed order, range, or monomial constraints, provided the selected prime variable has no arbitrary oscillating coefficient. It continues to hold after convolution with bounded sieve coefficients in other variables, with fixed divisor and logarithmic losses.*

*Proof.* Fix all but one prime variable and let $q$ be their product. The allowed values of the remaining prime lie in a bounded union of intervals inside $[M/q, 2M/q]$, subject to a lower bound $X^{\delta}$. Use Lemma 7.1 on that variable. Its error, multiplied by $q^{-\sigma}$, is $O(M^{1-\sigma} q^{-1} e^{-a\mathcal{L}(X)})$. The sum of $1/q$ over a fixed number of primes in $[X^{\delta}, O(X)]$ is bounded in terms of $\delta$ and the number of variables, by Mertens' estimate. Fixed logarithmic weights add only fixed logarithmic factors.

One can fix the other sieve variables first as well. Bounded sieve coefficients have harmonic absolute sums bounded by fixed logarithmic powers; grouping a fixed number of variables gives a fixed divisor bound. These factors are absorbed using (7). This statement does not assert cancellation for arbitrary coefficients on the prime variable itself. □

## 8 Earlier finite-sieve specializations

We use Li's finite decomposition at the short-interval exponent $2/43$. Its combinatorial regions and final numerical integrals are unchanged. The issue is the analytic error budget.

**Proposition 8.1** (accuracy parameter in the finite sieve)**.** *Fix a sufficiently small $\varepsilon > 0$ in the argument of [15], and put*

$$\eta = \frac{1}{2} X^{-41/43+\varepsilon}.$$

*For every fixed $D > 0$, there exist $X_0$ and $\kappa > 0$ such that*

$$\operatorname{meas}\left\{ x \in [X, 2X] : \pi(x+\eta x) - \pi(x) < \kappa \frac{\eta x}{\log X} \right\} \le X \mathcal{Q}_D(X)^{-1} \qquad (X \ge X_0). \tag{8}$$

*Here the finite region inequalities (17), (18), and (20) of [15] are used as external numerical inputs.*

*Proof.* Fix $\varepsilon$ first; in particular $\delta = \varepsilon^{1/3}$ is fixed. Whenever necessary, take $\varepsilon$ smaller so that the deterministic sieve approximation errors fit strictly inside Li's positive margin. No accuracy parameter below changes $\delta$, a moment order, or the number of Buchstab iterations.

Write $Q = \mathcal{Q}_D(X)$. We rerun the finite analytic estimates with an auxiliary accuracy $Q^{-R}$, where $R$ is a sufficiently large fixed constant. This is not an application of Li's theorem with a moving $B$. The following checks show directly that its proof allows the new accuracy.

**1. Pointwise input.** Replace the low-frequency cutoff by $T_0 = Q^{R_0+2}$ with a sufficiently large fixed $R_0$. Prime factors satisfy every fixed power of $Q^{-1}$ above that cutoff by Corollary 7.2 and Lemma 7.3. In particular, the saving $Q^{-R_0}$ remains available after the fixed logarithmic losses. Below it, keep the main integral in Lemma 7.1. The contribution of its remainder is bounded by a fixed power of $T_0$ and $\log X$ times $e^{-a\mathcal{L}(X)}$, and is therefore smaller than any prescribed fixed power of $Q^{-1}$.

**2. Mean and large values.** The estimates in Li's Lemmas 3.1–3.6 use fixed moments and fixed interpolations of classical mean-value and large-value inequalities. In Lemma 3.1 the integer moment satisfies $k \ll 1/\delta$, and the logarithmic loss has exponent $O(\delta^{-2})$; both are fixed. Every resulting case has one of two forms:

$$(\log X)^d Q^{-\alpha R_0} \quad \text{or} \quad (\log X)^d X^{-\nu},$$

where $d$ is fixed, $\alpha > 0$, and $\nu > 0$ depends on the fixed exponent margin. There are finitely many such cases. Thus the minimum relevant positive $\alpha$ can be chosen before $R_0$. By increasing $R_0$ and then $X$, every one is $O(Q^{-R})$ for any specified fixed $R$. The extra finite polynomial products in Lemmas 3.3 and 3.4 do not alter this conclusion. Lemmas 3.5 and 3.6 are regroupings and permutations of Lemma 3.2 and preserve the same accuracy accounting.

For completeness, one displayed interpolation in Lemma 3.2, case (1)(a), can be justified without its copied intermediate line. If the simultaneous large-value count $F$ is bounded by

$$V^{-6}M^3, \quad W^{-10}H^5, \quad U^{-4}K^2, \qquad MHK = X,$$

interpolate with weights 3/10, 1/5, and 1/2. Then

$$\begin{aligned} U^2V^2W^2F &\ll U^2V^2W^2(V^{-6}M^3)^{3/10}(W^{-10}H^5)^{1/5}(U^{-4}K^2)^{1/2} \\ &= V^{1/5}M^{9/10}HK. \end{aligned}$$

If $V \le M^{1/2}Q^{-R_0}$, this is at most $XQ^{-R_0/5}$. The positive power of the pointwise saving is all that is needed. This is the interpolation already present in Jia's Lemma 5, case I(a), printed page 28.

**3. The Watt-type estimates.** Li's Section 4 uses a fixed exponent margin in its mean-value estimates. The resulting error is a fixed power saving in $X$, or one of the small-factor cases already covered in step 2. Both dominate every required fixed power of $Q^{-1}$ by (7). The constants in these mean-value estimates depend on the fixed margins and coefficient bounds, not on a logarithmic accuracy exponent.

**4. Perron and the exceptional measure.** Here are the details behind the required error bookkeeping. For a relevant Dirichlet polynomial $F$, put

$$W_\eta(s) = \frac{(1+\eta)^s - 1}{s}.$$

On a dyadic range $|t| \asymp T$, $|W_\eta(b+it)| \ll \min(\eta, T^{-1})$, where $b = 1 + 1/\log X$. Opening the square, integrating $x^{s+\overline{s'}}$, and applying $2|ab| \le |a|^2 + |b|^2$, gives

$$\begin{aligned} &\int_X^{2X} \left| \int_{T_0 \le |t| \le X} F(b+it) W_\eta(b+it) x^{b+it}\, dt \right|^2 dx \\ &\qquad \ll X^3 (\log X)^3 \max_{T_0 \le T \le X} \min(\eta, T^{-1})^2 \int_{T \le |t| \le 2T} |F(b+it)|^2 dt. \end{aligned} \tag{9}$$

This is the estimate used in Jia's Lemmas 16–19 [16], whose Perron steps underlie Li's Section 5. If the right-hand maximum is $\ll \eta^2 Q^{-R}$, Chebyshev's inequality, with threshold $\eta X Q^{-r}$, yields exceptional measure

$$\ll X(\log X)^3 Q^{-R+2r}.$$

Choose $R > 2r + 10$ after all fixed decompositions. This leaves more than enough accuracy for (8). The low-frequency terms are evaluated as main terms, with errors $\ll \eta X Q^{-r}$ by step 1. Terms such as $\eta^2 X T_0^j$ and Perron truncation errors are harmless since $T_0^j = X^{o(1)}$ and $\eta X$ is a fixed positive power of $X$. Endpoint conventions contribute smaller errors as well. The same argument with the appropriate zeta factor handles the Type I terms.

**5. Separation of the finite constraints.** The number of prime variables and the number of defining inequalities are fixed. Standard dyadic partitions introduce only a fixed power of $\log X$. Alternatively, a relative mesh $Q^{-r_1}$ separates the variables into at most $Q^{O_\varepsilon(r_1)}$ boxes. Boundary strips in fixed monomial and ordering constraints have total tuple weight $\ll XQ^{-r_1}(\log X)^L + X^{1-\delta_0}(\log X)^L$ for fixed $L$ and $\delta_0 > 0$. The divisor short-interval estimate proved in Appendix E justifies this bound after all but one variable are fixed. Fubini and Markov transfer it to the short sums. Choose $r_1$ for the needed boundary accuracy first, and then choose $R$ large enough to cover the number of boxes. This avoids a circular choice of accuracy powers. For the lower bound one may also shrink the admissible polyhedral regions by a sufficiently small fixed amount: their boundary loss is absorbed in the fixed positive numerical margin.

**6. The sieve and its positive margin.** In Li's Lemma 5.6 the sieve level $X^{10^{-1000}}$ and the sifting threshold $X^\delta$ are fixed powers of $X$. The bounded sieve coefficients and the grouping of finitely many variables preserve fixed divisor bounds. The deterministic errors of size $O(\varepsilon)$ in the normalized main term need not become $Q^{-1}$: choose $\varepsilon$ small once. Only the fluctuating error and its exceptional set require the stronger accuracy.

In applying the small-factor estimates, a coefficient supported on rough integers is not automatically declared to have cancellation. The relevant groups contain prescribed prime factors, to which Lemma 7.3 applies after the remaining variables are fixed. If the group denoted $N$ in Li's Lemma 4.4 consists only of the tiny sieve variable, use the Type I estimate in Lemma 5.1 instead: the relevant range has $QR < X^{45/86}$ and $N < X^{41/344}$. Thus the proof does not need a pointwise estimate for arbitrary sieve weights.

Li's recorded numerical region estimates give the positive margin

$$1 - 0.03 - 0.992 + 0.027 = 0.005.$$

Choose the fixed deterministic errors and region-boundary losses less than half this margin. Steps 1–5 make all remaining errors smaller than a further fixed portion outside measure $XQ^{-1}$, for sufficiently large $X$. The same fixed sieve decomposition therefore gives (8), with a positive constant $\kappa$. The numerical inequalities themselves are precisely the external component specified in the proposition. □

*Remark* 8.2 (what is and is not imported). The standard statement $O_B(X/\log^B X)$ is not the imported quantitative conclusion. We use the finite algebraic estimates and the fixed positive sieve margin, and verify their accuracy dependence. The general free-parameter variance estimate in Matomäki–Radziwiłł–Shao–Tao–Teräväinen [17, Lemma 3.3] provides a separate modern formulation of the Perron/variance step. Their other results are not used to claim a new short-interval exponent.

## 8.1 The published-source alternative

**Proposition 8.3** (the published-source alternative)**.** *For sufficiently small fixed $\varepsilon > 0$, the counterpart of Proposition 8.1 holds with $\eta = \frac{1}{2}X^{-19/20+\varepsilon}$, using Jia's published proof in place of Li's refined decomposition.*

*Proof.* Jia fixes $\varepsilon$, $\varepsilon_1 = \varepsilon^2$ and $\delta = \varepsilon^{1/3}$ before the accuracy parameter. His mean-value arguments in Sections 2–5 have the same structure as steps 1–3 in Proposition 8.1: fixed moments, fixed powers of logarithms, and either a positive power of the pointwise saving or a fixed power saving in $X$. His Lemmas 16–19 are the Perron calculations quantified directly in (9). The subsequent fixed Buchstab regions, linear sieve levels and deterministic $O(\varepsilon)$ or $O(\delta)$ errors do not depend on the requested logarithmic accuracy. Thus the identical order of parameter choices gives an exceptional measure at most $X\mathcal{Q}_D(X)^{-1}$ for every fixed $D$. Jia's published positive numerical sieve margin supplies the constant $\kappa > 0$. We use that published margin as an external input just as in a normal application of his theorem. □

**Corollary 8.4** (intermediate Mills ranges)**.** *Using Jia's finite decomposition and published numerical margin, the four conclusions about* $\mathcal{W}(\boldsymbol{c})$ *hold for bounded sequences with* $\inf_n c_n > 20/19$*. Assuming the three numerical region inequalities (17), (18), and (20) of Li I, the corresponding lower bound is* $43/41$*.*

*Proof.* Choose the fixed $\varepsilon > 0$ in the appropriate proposition so that its interval exponent is less than $1 - 1/c_-$. The full-cell averaging argument in Section 4 applies also to the forward multiplicative windows here. Indeed, if $I = [a, b)$ has length $L \gg \eta X$, integrate over $[a, b/(1+\eta))$. Each prime is counted for at most $2\eta X$ units of endpoint measure. A cell with fewer than $\kappa L/(16 \log X)$ primes therefore contains at least $L/4$ measure of exceptional endpoints. Disjoint cells give the same bad-node bound, with arbitrary fixed accuracy, and the harmonic transfer proves the assertions. The numerical inputs of this corollary are independent of Hypothesis 1.3. □

## 9 Why the positive sieve margin alone is insufficient

Li III has the recorded lower-bound budget

$$1 - 0.02 - 0.03 - 0.97 + 0.04 = 0.02.$$

It is tempting to argue that continuity immediately permits a slightly smaller interval exponent. This misses a uniform analytic input.

Let $MH = X$, $H = X^h$, $M = X^{1-h}$, and put $b = 1 + 1/\log X$. On a one-separated large-value class write

$$V \le \sqrt{M}\,|M(b+it)| < 2V, \qquad W \le \sqrt{H}\,|H(b+it)| < 2W, \qquad R = \#\{\text{class}\}.$$

The classical mean-value and Halasz bounds [4] give, up to fixed powers of $\log X$,

$$R \ll \min\left\{\frac{M+T}{V^2}, \frac{M}{V^2} + \frac{MT}{V^6}, \frac{H^k+T}{W^{2k}}, \frac{H^k}{W^{2k}} + \frac{H^kT}{W^{6k}}\right\}. \tag{10}$$

The four-case argument in Li II [18, Lemma 3.1] bounds $V^2W^2R$ by, respectively,

$$X \cdot (\text{pointwise saving}), \quad M^{1/(2k)}T, \quad MH^{1/3}T^{1/(3k)}, \quad M^{1/(2k)}HT^{1-1/k}.$$

For intervals of length $X^\theta$, the last three are admissible when

$$h > 1 - 2k\theta, \qquad h > \frac{1-\theta}{2k}, \qquad h < \frac{1+2(k-1)\theta}{2k-1}. \tag{11}$$

At $\theta = 1/24$, the $k = 11$ and $k = 12$ ranges meet at $h = 1/12$. At $\theta = 1/24 - d$ a gap opens:

$$\frac{1}{12} - \frac{22d}{23} < h < \frac{1}{12} + 22d.$$

Reducing the outer cutoff $11/126$ by an infinitesimal amount does not remove this internal gap.

There is also a useful exponent witness. At $h = 1/12$, $T = X^{23/24+d}$, set

$$V = X^{11/48}, \quad W = X^{1/48-d/24}, \quad R = X^{1/2+d}.$$

Every individual bound in (10), for every positive integer $k$, permits these exponents, although $V^2W^2R = X^{1+11d/12}$. Adding the trivial mean value bound for $MH$ caps this at $X$: replace $R$ by $X^{1/2+d/12}$. Equality then remains possible at amplitudes far below their pointwise maxima, so no logarithmic saving follows from that additional trivial bound. This is an obstruction to these estimates, not an example of actual Dirichlet polynomials or a counterexample about primes.

## 10 An explicit modern two-factor bridge

Theorem 1.1 of Chen–Gupta–Li [2], with $q = q_1 = 1$, gives for $N \geq T^{2/3}$ and ordinary polynomial amplitude at least $U$,

$$R \ll T^{o(1)} \left( \frac{N^2}{U^2} + \frac{T^{1/2}N^3}{U^4} + \frac{TN^2}{U^4} + \frac{TN^{12/5}}{U^4} \right).$$

If $N \geq T^{5/6}$, the final two terms are dominated by the second. For a fixed divisor-bounded polynomial of length $M$, normalizing $U \asymp \sqrt{M}V$ and absorbing the coefficient bound into an arbitrarily small fixed power gives

$$R \ll_\xi X^\xi \left( \frac{M}{V^2} + \frac{MT^{1/2}}{V^4} \right) \qquad (\xi > 0,\ M \geq T^{5/6}). \tag{12}$$

The coefficient normalization is also detailed in [19, Lemma 2.3]. We use the underlying estimate, not an exponent extrapolation from its applications.

**Lemma 10.1** (two-factor estimate at 1/25)**.** *Fix $\delta, D > 0$, a desired accuracy $B_0 > 0$, and a divisor bound for the coefficients of $M(s)$. Let $MH = X$, $X^\delta \leq H \leq X^{11/126}$, and*

$$H(s) = \sum_{h \asymp H} \Lambda(h) h^{-s}, \quad Q = \exp[D(\log\log X)^2], \quad \eta = X^{-24/25}.$$

*There is a fixed $A > 0$ such that, uniformly for $Q^A \leq T \leq X$,*

$$\min(\eta, T^{-1})^2 \int_T^{2T} |M(b+it)H(b+it)|^2 \, dt \ll \eta^2 Q^{-B_0}.$$

*Proof.* The quantitative prime-polynomial estimate in Lemma 7.1 gives, for every fixed $A_1$, a bound

$$|H(b+it)| \ll Q^{-A_1} \qquad (Q^{A_1+2} \leq |t| \leq 2X).$$

Indeed the uniform error is $\exp[-a(\log X)^{1/3}/(\log\log X)^{1/3}]$, smaller than every fixed power of $Q^{-1}$, and the retained main integral is $O(1/|t|)$. The difference kernel $(v^w - u^w)/w$ removes an artificial pole at $w = 0$ in the Perron argument. Hence $W \ll \sqrt{H}Q^{-A_1}$ and $V \ll \sqrt{M}(\log X)^C$. Standard separation and dyadic amplitude classes cost fixed logarithmic powers. It suffices to bound $\min(\eta, T^{-1})^2 V^2 W^2 R/X$.

First suppose $\delta \leq h \leq 81/1000$ and take $k = \lfloor 1/h \rfloor$. Then $k \geq 12$. The normalized exponents in the three non-diagonal classical cases are

$$e_b = \frac{1-h}{2k} - \frac{1}{25}, \quad e_c = -\frac{2h}{3} + \frac{24}{75k}, \quad e_d = \frac{(2k-1)h - 1 - (2k-2)/25}{2k}.$$

They satisfy

$$e_b \leq -\frac{1}{650}, \qquad e_c \leq -\frac{8\delta}{25},$$

and

$$e_d \leq \begin{cases} -17/24000, & k = 12, \\ -6/4225, & k \geq 13. \end{cases}$$

The diagonal case retains the pointwise $Q^{-A_1}$ saving.

It remains to cover $81/1000 \leq h \leq 11/126$. Here $M \geq X^{115/126} \geq T^{5/6}$ for $T \leq X$. Use $k = 11$ and classify terms as follows. If $T \leq M$ or $V \geq T^{1/4}$, the classical bounds give $R \ll MV^{-2}(\log X)^C$. Otherwise (12) and the classical mean value bound give

$$R \ll X^\xi MT^{1/2}V^{-4}, \qquad R \ll TV^{-2}(\log X)^C.$$

Similarly, if $T \le H^k$ or $W \ge T^{1/(4k)}$, then $R \ll H^k W^{-2k} (\log X)^C$. Otherwise both $R \ll TW^{-2k}(\log X)^C$ and $R \ll H^k T W^{-6k}(\log X)^C$ hold.

If both factors are in their diagonal regimes, interpolation with weights $1 - 1/(2k)$ and $1/(2k)$ gives

$$V^2 W^2 R \ll V^{1/k} W M^{1-1/(2k)} H^{1/2} (\log X)^C \ll X Q^{-A_1} (\log X)^C.$$

In particular, this case does not incur an $X^\xi$ loss. The other three cases give, with logarithmic factors suppressed,

$$X^{\xi/k} M^{1/k} T^{1-1/(2k)}, \qquad M H^{1/3} T^{1/(3k)}, \qquad X^{\xi/k} M^{1/k} H T^{1-3/(2k)}.$$

For the first and third expressions, use the interpolation weights $1 - 2/k, 1/k, 1/k$. Since $\min(\eta, T^{-1})^2 T^a \le \eta^{2-a}$ for $0 \le a \le 2$, their normalized exponents, before the arbitrarily small $\xi$ loss, are

$$e'_b = \frac{1 - 2h - (2k-1)/25}{2k}, \quad e'_c = -\frac{2h}{3} + \frac{24}{75k}, \quad e'_d = \frac{(2k-2)h - 1 - (2k-3)/25}{2k}.$$

Uniformly on the stated patch these obey

$$e'_b \le -\frac{1}{11000}, \qquad e'_c \le -\frac{137}{5500}, \qquad e'_d \le -\frac{1}{1575}. \tag{13}$$

Take $\xi < 1/44000$. All non-diagonal cases retain a fixed power saving. Because $Q = X^{o(1)}$, these powers dominate $Q^{-B_0}$. Finally choose $A_1$ large enough for the diagonal case and its fixed logarithmic losses. □

*Remark* 10.2. The new input repairs a whole interval of factor lengths, not just one exceptional amplitude tuple. The classical $k = 12$ upper range and the modern $k = 11$ lower range overlap once

$$\frac{1 + 22\theta}{23} > \frac{1 - 21\theta}{2}, \quad \text{equivalently} \quad \theta > \frac{21}{527}.$$

This indicates further room in this particular lemma. For the full sieve we only need the explicitly proved $1/25$ version.

## 11 Exact trilinear certificates

Let $M = X^m$, $H = X^h$, $K = X^{1-m-h}$ and $T = X^{1-\theta}$. The additional regions needed from the Harman-type input are

$$\begin{aligned}
&\mathcal{H}_1(\theta) : 2\theta < h \le m < \tfrac{1}{2}, \quad 2h - 2\theta < m < 4\theta;\\
&\mathcal{H}_2(\theta) : 2\theta < h < 4\theta, \quad |4m + 3h - 2| < 2\theta, \quad h \le m < \tfrac{1}{2};\\
&\mathcal{H}_3(\theta) : 2\theta < h < 4\theta, \quad |4m + h - 2| < 2\theta, \quad h \le m < \tfrac{1}{2}.
\end{aligned}$$

In $\mathcal{H}_1$ the factor $M$ must supply the prime-polynomial pointwise saving; in the other two regions it is $H$. We use compact interiors of these regions, so equalities along geometric boundaries may be omitted without changing the integrals.

Write the normalized amplitude deficits as

$$|M(1/2 + it)| \asymp X^{m/2-a}, \quad |H(1/2 + it)| \asymp X^{h/2-b}, \quad |K(1/2 + it)| \asymp X^{(1-m-h)/2-c},$$

and let the large-value cardinality have exponent $f$. For a monomial $M^r H^s K^u$, put

$$L = rm + sh + u(1 - m - h), \qquad d = ra + sb + uc.$$

Classical estimates give

$$f \le \max(2d, 2d + 1 - \theta - L), \qquad f \le \max(2d, 6d + 1 - \theta - 2L), \tag{14}$$

up to fixed logarithmic factors. These bounds include mixed products; individual powers alone need not give the desired conclusion.

**Lemma 11.1** (rationally certified regions)**.** *Uniformly for $\theta \in [1/25, 1/24]$, the inequalities (14) for the finitely many monomials in the attached certificates imply*

$$f - 2(a+b+c) \le -\frac{z}{100}. \tag{15}$$

*Here $z$ is the minimum of the saving factor's amplitude deficit and the positive slacks in the defining region inequalities. Consequently, on every compact interior of any $\mathcal{H}_i$, the corresponding trilinear mean square has arbitrary fixed $Q^{-1}$ accuracy whenever the indicated prime factor does.*

*Proof.* Each maximum in (14) is a disjunction of two affine inequalities. The proof certificates use rational coefficients in the eight variables

$$(m, h, a, b, c, f, z, \theta).$$

They cover all disjunctions by a finite binary tree. At each leaf, nonnegative rational multiples of the allowed inequalities sum exactly to $f - 2a - 2b - 2c + z/100 \le 0$. The two certificates have the following sizes:

| region | leaves | maximum depth | maximum sum of dual weights |
|---|---|---|---|
| $\mathcal{H}_1$ | 70 | 17 | 1521/400 |
| $\mathcal{H}_2$ | 36 | 9 | 573/200 |

The verifier uses only Python's integer arithmetic, `Fraction`, and JSON parsing. It checks the provenance of every inequality, the exact extremal lengths on the region vertices, complete tree coverage, nonnegativity of every dual coefficient, exact equality of coefficient vectors, and nonpositivity of the resulting constant. It does not import the numerical optimizer that found the trees.

For $\mathcal{H}_3$, replace $M$ by $K$:

$$|4(1-m-h) + 3h - 2| = |4m + h - 2|.$$

The required order $h \le 1 - m - h < 1/2$ holds in this range ($\theta \le 1/15$ suffices).

To interpret the certificate analytically, first discard classes where a normalized amplitude is less than 1: the ordinary second moment for the product of the other factors gives a fixed power saving. On the other classes, all deficits lie between 0 and half the corresponding length exponent, apart from harmless logarithmic normalizations. On a fixed compact interior, let the region slack be at least $\rho$. If the saving factor has deficit less than $\rho$, (15) retains a fixed positive power of its pointwise $Q^{-A}$ saving. If its deficit is at least $\rho$, the certificate gives the fixed power saving $X^{-\rho/100}$. All logarithmic errors in the input inequalities are multiplied by bounded sums of rational dual weights. Taking $A$ large covers these errors and the finitely many amplitude classes. This proves the $Q$-accuracy conclusion. □

## 12 A short-to-long comparison with a free accuracy

The extra trilinear input is compared to a longer interval. We give the comparison explicitly, in particular retaining the cancellation of the low-frequency kernels. For a coefficient sequence of fixed divisor bound, let

$$F(s) = \sum_{n \asymp X} a_n n^{-s}, \qquad S_h(x) = \sum_{x - hx < n \le x} a_n,$$

and put

$$\eta_1 = e^{-3(\log X)^{1/3}}, \qquad T_0 = e^{(\log X)^{1/3}}, \qquad T = X^{\varepsilon}/\eta.$$

Suppose that

$$\int_{T_0}^{T} |F(1/2+it)|^2\,dt \ll XQ^{-R}.$$

Then, for every fixed $r$, increasing $R$ as necessary gives

$$S_\eta(x) = \frac{\eta}{\eta_1} S_{\eta_1}(x) + O(\eta X Q^{-r}) \tag{16}$$

outside measure $O(XQ^{-r})$.

To see this, use the Mellin kernels $k_h(s) = (1-(1-h)^s)/s$. For $|t| \le T_0$,

$$\left| k_\eta(s) - \frac{\eta}{\eta_1} k_{\eta_1}(s) \right| \ll \eta\eta_1(1+|t|).$$

The low-frequency contribution is therefore at most

$$\eta X \eta_1 T_0^2 (\log X)^C = \eta X e^{-(\log X)^{1/3}} (\log X)^C,$$

smaller than every fixed power of $Q^{-1}$ relative to $\eta X$. Mellin–Plancherel controls the middle frequencies by $O(\eta^2 X^3 Q^{-R})$ in mean square over $x \in [X, 2X]$. For the tail, the classical bound

$$\int_U^{2U} |F(1/2+it)|^2 dt \ll (U+X)(\log X)^C$$

and kernel decay $O(U^{-1})$ give a mean square

$$\ll X^2(1/T + X/T^2)(\log X)^C,$$

which has a fixed power saving relative to $\eta^2 X^3$. Chebyshev's inequality proves (16). Thus no multiplication of a weak pointwise error by the exponentially large cutoff $T_0$ is required. The correct scaling is $\eta/\eta_1$; the occurrence of $\eta/\delta$ in the transcribed comparison lemma in Li II/III is not used.

## 13 Stability of the signed finite sieve

Keep the initial cutoff

$$a_0 = \frac{11}{126}$$

fixed. This is possible because Lemma 10.1 retains that cutoff throughout an interval of short-interval exponents below $1/24$. The original decomposition begins with

$$S(A, (2X)^{1/2}) = S_1 - S_2 + S_3, \qquad S_3 = \sum_{(t_1,t_2)\in\Omega} S(A_{p_1p_2}, p_2),$$

where all summands of $S_3$ are nonnegative. The limiting exponent domain, after the usual fixed-factor endpoint conventions, is

$$\Omega = \{a_0 \le t_2 \le t_1 \le 1/2, \quad t_1 + 2t_2 \le 1\}.$$

**Lemma 13.1** (finite signed sieve stability)**.** *Suppose a finite Buchstab decomposition has a strictly positive normalized lower-bound margin at $\theta_0$. Assume its initial signed terms remain uniformly evaluable in a neighborhood below $\theta_0$, and its other evaluations use finitely many continuously moving affine exponent regions. For negative children, assume that complete closed interval coverings remain available, and that a reserved positive interval-exponent slack makes the estimates valid on those closed coverings. For freely selected good terms, compact interior estimates suffice. Assume also that the universally quantified child ranges have no parameter-only zero constraint or coincident moving boundary that creates a positive-measure discontinuity at $\theta_0$. If all discarded comparator densities are integrable, the same scheme has positive margin for some $\theta < \theta_0$.*

*Proof.* Remove thin boundary strips from the positive parent regions before applying any further signed decomposition. The removed parent sums are nonnegative, so discarding them preserves the desired lower bound. Within each retained parent region, repeat the exact Buchstab identities, protecting all negative children by their complete coverings and the reserved exponent slack. Do not erode individual members of a closed interval covering and create gaps at shared endpoints. For the terminal positive terms, use the eroded good regions and discard the others.

The corresponding comparator sums satisfy the same identities. Their total additional normalized deficiency is bounded by integrals over the removed strips and the symmetric differences of the terminal regions. These tend to zero as the strip thickness tends to zero and the parameter tends to $\theta_0$. Integrability gives dominated convergence. The hypothesis concerning child ranges ensures that removing parent boundary strips is sufficient; an internal disappearing coverage band of positive parent measure would violate it. Choose the strip thickness to consume less than one quarter of the original margin, then move $\theta$ by a positive amount small enough to retain every strict analytic inequality and consume less than another quarter. The remaining margin is positive. □

We apply this lemma to the actual finite decomposition, rather than postulating continuity of the published theorem. Here are the checks specific to Li III.

**Initial signed terms.** The initial $S_1$ and $S_2$ use the Type I sieve input, with $t_1 \le 1/2 + o(1)$, and the small-prime terminal evaluation. The Type I inequalities have strict room at these lengths; for instance $2t_1 \le 1 < 1+\theta$. The terminal prime factors below $X^{a_0}$ are covered uniformly by Lemma 10.1. This is precisely where the old integer-moment argument would fail. The auxiliary small sieve exponent (written $10^{-1000}$ in the source) may be chosen sufficiently small after the retained region margins have been fixed. Its value and all sieve depths then remain fixed as $X$ grows.

**The remaining analytic regions.** Li III, Lemma 3.2, uses seven fixed moment patterns. We verified all $7 \cdot 8 = 56$ diagonal/off-diagonal branches by exact nonnegative rational Hölder identities; the patterns and complete moving regions are recorded in Appendix A. Two faces must move with the auxiliary exponent:

$$\mathcal{C}_5 : \quad m + h \ge 1 - 12\theta, \qquad \mathcal{C}_7 : \quad m \le 12\theta.$$

Keeping the printed halves fixed would give an incorrect perturbation. All non-pointwise monomials contain a strictly positive power of $T$. Thus closed length faces at an auxiliary exponent $\theta_a$ acquire a fixed power saving when the actual interval exponent is $\theta_a + e$, $e > 0$. Diagonal cases retain a positive fixed power of the prime-polynomial saving. The Watt/Type I faces are

$$2m + h \le 1 + \theta, \quad 4m + 6h \le 3 + \theta, \quad 4h \le 1 + 3\theta,$$

or $m \le (1+\theta)/2$, $h \le (1-\theta)/8$. They likewise allow a fixed positive height slack; their harmless $T^{\varepsilon}$ loss is chosen smaller than that slack. The additional trilinear regions are supplied independently by Lemma 11.1.

**The actual negative-child ranges.** For $U_{0201}, \dots, U_{0209}$, put $s = t_1 + t_2$ and $h = t_3$. The parent conditions give

$$\frac{139}{336} \le s \le \frac{503}{1008}, \qquad h \ge a_0.$$

Consequently

$$s + h - \frac{1}{2} \ge \frac{1}{1008}, \qquad \frac{1}{2} - s \ge \frac{1}{1008}. \tag{17}$$

Both moving half-faces therefore remain valid if $\theta_a = 1/24 - d$ and $0 < d < 1/12096$. $U_{0210}, \dots, U_{0212}$ use only $\mathcal{C}_1, \mathcal{C}_2, \mathcal{C}_3$, with $m = t_1$, $h = t_2 + t_3$; the two corrected faces do not occur there. For $U_{0303}$, $m = s \le 11/24$ gives a $1/24$ margin from the upper half-face, and $s + h \ge 115/264 + 11/126 > 1/2$.

The full $U_{02}$ coverage uses the union of the classical regions and the Harman strip regions; it is not a classical-only covering. To use the closed strip endpoints rigorously, reserve $0 < e < d$ and put

$$\theta' = \theta_a + e/2 < 1/24, \qquad \theta_{\text{actual}} = \theta_a + e < 1/24.$$

The region $\mathcal{H}_1$ is impossible here because $m = s > 4\theta'$. For $\mathcal{H}_2$ or $\mathcal{H}_3$ closed at $\theta_a$, the same point is in a compact interior at $\theta'$: its upper $h$ margin is at least $2e$, its two strip margins are at least $e$, and

$$h - 2\theta' \ge a_0 - \frac{1}{12} = \frac{1}{252}.$$

For $\mathcal{H}_2$, $m - h \ge 83/336$ and $1/2 - m \ge 1/1008$. For the $\mathcal{H}_3$ swap $m' = 1 - s - h$, one has $m' - h \ge 169/1008$ and $1/2 - m' \ge 1/1008$. The truncation height $X^{e/4}/\eta \le X^{1-\theta'}$ for large $X$ fits Lemma 11.1 and leaves a fixed power saving in the transfer tail. Thus shared negative-child endpoints are retained without gaps.

The other negative children use the following uniform Type I groupings. In $U_{0301}$ use $(m, h) = (t_1 + t_2, t_3)$, and in $U_{0302}$ use $(m, h) = (t_2 + t_3, t_1)$; the worst endpoint is $t_3 = t_2$ in both cases. In $U_{0401}, U_{0402}$ the reversed grouping is $(m, h) = (1 - t_1 - t_2, t_2)$, independent of $t_3$. The defining parent conditions are precisely sufficient for these full ranges. The combination of $\mathcal{C}_7$ and Type I controls $U_{0303}$.

**Finite boundary geometry.** The primitive classical forms have integer left-hand coefficients of absolute value at most 82 and right-hand $\theta$-slopes at most 42. Substituting the finitely many groupings of $t_1, t_2, t_3, 1 - t_1 - t_2 - t_3$ gives affine child endpoints. Their order can change only on finitely many affine parent faces. An exact rational check of the 3209 substituted forms finds no parameter-only zero face; coincident moving faces at $1/24$ are the half-face families just handled by (17) and the $U_{0303}$ margins. Every other nonidentity endpoint comparison has nonzero parent dependence and can be protected by removing a parent boundary strip. The next distinct parallel-face contact below $1/24$ is at $1/25$. These checks supplement the complete analytic case proofs and the actual negative-child analysis; a collision scan alone would not prove sieve stability.

**Signed pruning and numerical margin.** Whole positive parents may be pruned before their signed refinements. The relevant exact identities include

$$S_{303} = S_{3031} - S_{3032} - S_{3033} + S_{3034} + S_{3035},$$
$$S_{304} = S_{3041} - S_{3042} - S_{30431} + S_{30432} + S_{30433}.$$

Shrinking the "good" predicate enlarges the negative terms $S_{3033}$ and $S_{30431}$, so their full-range Type I evaluations are essential. Only the positive terminal failures $S_{3035}$ and $S_{30433}$ are discarded. For the reverse-Buchstab recovery, retain the old bad-parent domain and compact subsets of its new good triples; this reduces a positive saving by a controlled comparator integral. No short-interval upper bound on a removed boundary strip is asserted. The cost is charged solely to the same identity on the long comparator.

After reversal, the $\beta$ coefficient is the exact indicator of $p_3$-rough integers, rather than an arbitrary divisor-bounded sequence. Its unique ordered factorization has a bounded number of primes, all at least $X^{a_0}$. Condition on all but one of those primes and apply the uniform prime-polynomial bound on its remaining interval. The outer reciprocal-prime sums are bounded. This supplies the required quantitative prime-factored bound even when a group consists only of $\beta$; see Appendix C.

All prime exponent variables in the four integrals are at least $a_0$. The respective densities are bounded by $a_0^{-3}, a_0^{-5}, a_0^{-5}, a_0^{-4}$, hence by

$$a_0^{-5} = \frac{31757969376}{161051} < 200000.$$

Their boundary losses tend to zero by dominated convergence. Using the four recorded numerical inequalities

$$\mathrm{loss}_1 < 0.02, \quad \mathrm{loss}_2 < 0.03, \quad \mathrm{loss}_3 < 0.97, \quad \mathrm{recovery} > 0.04$$

leaves positive room to perform this pruning.

The fully enumerated budget in Appendix D permits

$$\rho = 10^{-20}, \qquad d = 2 \cdot 10^{-23}, \qquad e = 10^{-23}.$$

Here $\rho$ is the parent/terminal erosion width, the analytic length regions use $\theta_a = 1/24 - d$, and the actual interval exponent is $\theta_a + e$. The total comparator cost is at most $0.00295856 < 0.005$. Also $84d < \rho/4$ and $d < \min(1/600, 1/12096)$, so every retained endpoint-order sign and all fixed half-face gaps persist. There is a remaining normalized margin greater than 0.017 before the fixed deterministic sieve errors, which are chosen smaller than that margin. Thus the complete lower-bound sieve works at

$$\theta_* = \frac{1}{24} - 10^{-23}.$$

All parameters, region erosions, sieve levels, and moment orders are fixed before the requested exceptional accuracy is chosen. The arguments of Lemmas 10.1 and 11.1 and (16) supply arbitrary fixed powers of $Q^{-1}$. The remaining Perron and Chebyshev steps lose only fixed powers of $Q$ or $\log X$. Thus the exceptional measure is $X \exp[-D(\log\log X)^2]$ for any prescribed fixed $D$. For the fixed-length formulation in Theorem 1.1, take $\eta = \frac{1}{2}X^{\theta_*-1}$. Then $X^{\theta_*}/2 \le \eta x \le X^{\theta_*}$ for $x \in [X, 2X]$. The prime count in the shorter multiplicative window gives the claimed fixed-window lower bound after decreasing $\kappa$. All moment and truncation arguments allow these fixed factors. This proves the analytic assertion of Theorem 1.1.

*Remark* 13.2 (source convention and independent numerical check). Our numerical hypotheses are stated for the literal printed regions, including the empty $V_0$ with $t_1 < 1/16$. Correcting this cap to $1/6$, as suggested by Lemma 5.2, is also safe under those same hypotheses. Indeed the added region has $t_1 \le 1/6$, $t_2 \le 1/8$ and lies inside the raw $U_{0301}$ region:

$$2t_1 + 3t_2 \le 17/24 < 25/24, \quad 4t_1 + 10t_2 \le 23/12 < 73/24, \quad 4t_2 \le 1/2 < 9/8.$$

Thus the union $U_0$ and its loss integral are unchanged. Moving these positive parents into $V$ and enlarging the good predicate can only reduce the first two losses and increase the recovery. This is a monotonicity argument, not an assumption about which convention the author used in an unpublished calculation.

An independent exploratory quadrature gives

$$I_0 \approx 0.966495918091412$$

for the third loss under either convention. Gauss orders $12, 24, 48$ agree at the displayed precision; the source's conservative Buchstab upper envelope gives approximately $0.967062500408101$. These computations are not interval-certified enclosures and do not replace the explicit external numerical hypotheses. The other three integrals have not been independently recomputed in this note.

*Completion of Theorem 1.1.* The signed sieve argument proves (1) under Hypothesis 1.3. For $c_- > c_*$ one has $\theta_* < 1 - 1/c_-$. The transfer theorem of Section 4 therefore applies to every sequence in the stated range. The strict nesting argument gives exact floors, and the measure/category proposition supplies the last two conclusions. Finally, (2) proves admissibility of $c_{\mathrm{new}}$; $24/23 > c_*$ proves the other stated endpoint. □

## 14 What remains to reach 25/24

The new two-factor estimate and the independently certified trilinear regions both work at $\theta = 1/25$. This does not imply that the full signed sieve has positive margin there. To obtain the full interval exponent $1/25$ and then the range $c > 25/24$, one must quantitatively optimize the remaining admissible regions and certify their final Buchstab deficiency below 1.

The present conclusion is strictly stronger than importing the statement at $1/24 + \varepsilon$: it supplies an actual positive movement below $1/24$ and includes $c = 24/23 - 10^{-23}$. The explicit decrement is a conservative consequence of the checked finite-geometry budget, rather than a numerical optimization of the sieve. The embedded verification materials contain the exact rational certificates and an independent verifier. The single LaTeX source embeds those materials so that the mathematical computation can be reproduced without a floating-point optimizer.

## A The seven classical moving regions

Put $\tau = 1 - \theta$, $k = 1 - m - h$. For a separated value class, $V, W, U$ denote the normalized amplitudes of $M, H, K$. For a product polynomial $P = M^r H^s K^u$, let its length be $L = M^r H^s K^u$ and its amplitude be $A = V^r W^s U^u$. The classical bounds split exhaustively into a diagonal row $R \ll LA^{-2}$ or two simultaneous off-diagonal rows

$$R \ll TA^{-2}, \qquad R \ll LTA^{-6},$$

with fixed logarithmic factors understood. The following three product choices give eight branches each.

| Region | Product polynomials | Pointwise saving factor |
|---|---|---|
| $\mathcal{C}_1$ | $M^3, H^5, K^2$ | $\lvert M(b+it)\rvert^{1/5}$ |
| $\mathcal{C}_2$ | $M^3, H^4, K^2$ | $\lvert M(b+it)\rvert^{1/2}$ |
| $\mathcal{C}_3$ | $M^2, H^5, K^3$ | $\lvert H(b+it)\rvert^{1/3}$ |
| $\mathcal{C}_4$ | $M^2, H^4, K^3$ | $\lvert H(b+it)\rvert^{2/3}$ |
| $\mathcal{C}_5$ | $M^2H, H^6, K^2$ | $\lvert H(b+it)\rvert$ |
| $\mathcal{C}_6$ | $M^2H, H^7, K^2$ | $\lvert H(b+it)\rvert$ |
| $\mathcal{C}_7$ | $M^2, H^6, K^2H$ | $\lvert H(b+it)\rvert$ |

The embedded `verify_threefactor_fraction.py` contains all 56 target monomials and their rational Hölder weights. It reconstructs the valid rows, checks nonnegative weights summing to one, verifies every exponent exactly, and checks that every $T$-power is in $[0, 1]$. The pointwise cases are $X$ times the positive powers in the table; all other branches have a positive $T$-power and give precisely the length conditions below.

For example, branch OOD of $\mathcal{C}_5$ uses weights $11/24, 1/72, 1/36, 1/2$ on the first product's mean row, its Halász row, the second product's Halász row, and the third product's diagonal row. Their sum is one and they give

$$V^2W^2U^2R \ll T^{1/2}M^{1/36}H^{13/72}K(\log X)^C.$$

Putting $T = X^{1-\theta}$ yields $70m + 59h \geq 36\tau$.

With $m, h, k > 0$, the following closed inequalities suffice at the auxiliary exponent. The actual exponent has the reserved positive slack.

$$
\begin{aligned}
\mathcal{C}_1:\quad & m+h\le (15+14\theta)/29, \qquad h\ge 2\tau/11,\\
& 29m-h\le 9+20\theta, \qquad m\ge 6\tau/19,\\
& -m+29h\le 5+24\theta, \qquad 12m+11h\ge 6\tau.\\
\mathcal{C}_2:\quad & m+h\le (6+5\theta)/11, \qquad h\ge \tau/5,\\
& 6m+h\le 2+4\theta, \qquad 2\tau/7\le m\le 8\theta,\\
& m+8h\le 2+6\theta, \qquad 6m+5h\ge 3\tau.\\
\mathcal{C}_3:\quad & m+h\le (13+6\theta)/19, \qquad h\ge 2\tau/11,\\
& 12m+h\le 6+6\theta, \qquad m\ge 14\tau/29,\\
& -m+19h\le 3+16\theta, \qquad 12m+11h\ge 8\tau.\\
\mathcal{C}_4:\quad & 1-8\theta\le m+h\le (5+2\theta)/7, \qquad h\ge \tau/5,\\
& 6m+h\le 3+3\theta, \qquad m\ge 5\tau/11,\\
& -m+7h\le 1+6\theta, \qquad 6m+5h\ge 4\tau.\\
\mathcal{C}_5:\quad & 1-12\theta\le m+h\le (6+5\theta)/11,\\
& 35m+23h\le 17+18\theta, \qquad 2m+h\ge 10\tau/11,\\
& 2m+13h\le 2+22\theta, \qquad 70m+59h\ge 36\tau.\\
\mathcal{C}_6:\quad & m+h\le (7+6\theta)/13, \qquad 41m+27h\le 20+21\theta,\\
& 2m+h\ge 12\tau/13, \qquad 2m+15h\le 2+26\theta,\\
& 82m+69h\ge 42\tau.\\
\mathcal{C}_7:\quad & 2m+h\le (12+10\theta)/11, \qquad 70m+11h\le 34+36\theta,\\
& 5\tau/11\le m\le 12\theta, \qquad m+6h\le 1+11\theta,\\
& 35m+12h\ge 18\tau.
\end{aligned}
$$

Three all-off-diagonal conditions are redundant on these regions. For $\mathcal{C}_1$, the slack in $m\le 20\theta$ is at least $(6168\theta-107)/319\ge 3493/7975$. For $\mathcal{C}_3$, the condition $m+h\ge 1-20\theta$ has the same lower bound on its slack. For $\mathcal{C}_6$, the condition $m+h\ge 1-14\theta$ has slack at least $(176\theta-7)/13\ge 1/325$. These assertions hold throughout $[1/25,1/24]$.

Every non-pointwise branch is a monomial $T^{\alpha}M^{r}H^{s}K^{u}$ with $0<\alpha\le 1$. Reducing the critical height by $X^{e}$ gives a power saving $X^{-\alpha e}$ even on a closed length face. For arbitrary $T$, the factor $\min(\eta,T^{-1})^{2}T^{\alpha}$ is largest relative to $\eta^{2}$ at $T=1/\eta$. This proves the full weighted mean-square assertion, not merely its value at one height.

## B Direct verification of the moving Type I faces

Let $MHL=X$ and $L(s)$ be the unweighted zeta-type polynomial. Watt's estimate [20], in the form stated in [3, Lemma 4.1], is

$$\int_{T}^{2T}|L(1/2+it)|^{4}|N(1/2+it)|^{2}dt\ll (T+N^{2}T^{1/2}+NL^{2}T^{-2})X^{\xi}$$

for arbitrarily small fixed $\xi>0$. For the first Type I region, Cauchy's inequality with the second moment of $M^{2}H$ and Watt applied to $N=H$ gives

$$\int_{T}^{2T}|MHL|^{2}dt\ll (M^{2}H+T)^{1/2}(T+H^{2}T^{1/2}+HL^{2}T^{-2})^{1/2}X^{\xi}.$$

The first four terms are

$$MH^{1/2}T^{1/2}, \quad MH^{3/2}T^{1/4}, \quad T, \quad HT^{3/4}.$$

At $T = X^{1-\theta}$ these are at most $X$ under exactly the three Type I inequalities displayed in Section 13, with the condition $\theta > 0$ for the third term. The last Watt term contributes at most $2X/\sqrt{T}$. The length faces give $\log_X L \geq 65/192$; since this input is used only for $T \geq \sqrt{L}$, the last term has a uniform power saving.

For the second Type I region, use the second moment of $M^2$ and Watt with $N = H^2$. The first four terms become

$$MT^{1/2}, \quad MH^2T^{1/4}, \quad T, \quad H^2T^{3/4}.$$

They are at most $X$ if $m \leq (1+\theta)/2$ and $h \leq (1-\theta)/8$. The last Watt term is again at most $2X/\sqrt{T}$, and now $\log_X L \geq 23/64$. The potentially saturated terms have $T$-powers at least $1/4$. On closed faces at $\theta_a$, an actual interval exponent $\theta_a + e$ therefore gains at least $X^{-e/4}$. Choosing $\xi < e/8$ preserves a fixed power saving. This directly derives the moving Type I faces; it does not interpolate between two published endpoint statements.

## C The rough factor after reversal

Let $B \geq X^{a_0}$, $z \geq X^{a_0}$, and

$$F_{B,z}(s) = \sum_{\substack{B<n\leq 2B \\ p|n \Rightarrow p \geq z}} n^{-s}, \qquad B \leq O(X).$$

For $1/2 \leq \sigma \leq 1 + 1/\log X$ and $|t| \leq 2X$, the uniform prime-polynomial estimate gives

$$|F_{B,z}(\sigma+it)| \ll_{a_0} \frac{B^{1-\sigma}}{\log X}\left((1+|t|)^{-1} + \exp\left[-c\frac{(\log X)^{1/3}}{(\log\log X)^{1/3}}\right]\right).$$

To prove this, write each rough integer uniquely as an ordered product $n = p_1 \cdots p_r$, $z \leq p_1 \leq \cdots \leq p_r$. The number $r$ is bounded in terms of $a_0$. Fixing $p_2, \ldots, p_r$ and writing their product as $q$, the variable $p_1$ lies in $(B/q, 2B/q] \cap [z, p_2]$ (with the evident $r = 1$ convention). Apply the uniform prime estimate on that interval. Its contribution is bounded by $B^{1-\sigma}/q$ times the common error factor and $1/\log X$. The sum of $1/q$ is bounded by a fixed power of $\sum_{X^{a_0} \leq p \leq O(X)} 1/p = O_{a_0}(1)$. Repeated primes are included once by the ordered factorization. Finite order and monomial restrictions leave bounded unions of intervals after conditioning and are handled in the same way.

Thus for $|t| \geq Q^A$ this rough indicator has an arbitrary fixed $Q^{-1}$ saving after increasing $A$. The assertion relies on its exact rough-indicator coefficients. It is not asserted for an arbitrary rough-supported coefficient sequence. This distinction is needed in the reversed $S_{304}$ terms.

## D An explicit decrement from the finite geometry

We now make the continuity argument quantitative. Throughout this construction the printed $V_0$ remains empty. Its unused $\mathcal{H}_1$ faces may be included in the inventory as an overestimate; no new positive-volume good region is silently activated.

For the universally quantified negative children, assign each of $t_1, t_2, t_3, 1-t_1-t_2-t_3$ to $M, H$, or neither, with both groups nonempty. There are $3^4 - 2 \cdot 2^4 + 1 = 50$ assignments. The 64 primitive faces and nine fixed geometry faces give $64 \cdot 50 + 9 = 3209$ forms. Eliminating the single child variable $t_3$ only requires comparisons between its affine endpoints and parent-only face tests. Thus at most $3209^2 + 3209$ events suffice.

The ordinary good predicates use four exponent tuples:

$$(t_1, t_2), \quad (t_1, t_2, t_3), \quad (t_1, t_2, t_3, t_4), \quad (1 - t_1 - t_2 - t_3, t_2, t_3, t_4).$$

Their respective nonempty group assignment counts are $2, 12, 50, 50$, so they contribute $64(2+12+50+50) = 7296$ faces. The 27 original parent components have at most 12 faces each including the common domain conditions. There are fewer than 40 additional fixed integration-domain faces. Boolean unions and exclusions create no additional affine boundary. Consequently the complete event count obeys

$$N \leq 3209^2 + 3209 + 7296 + 324 + 40 = 10\,308\,550 < 11\,000\,000.$$

The exact inventory, after partial deduplication, has 496 587 events; the proof uses the larger bound $N = 11\,000\,000$.

Write a primitive face as $A \cdot t \leq b_0 + b_1\theta$. Its integer coefficients have magnitude at most 82, and $|b_1| \leq 42$. A child endpoint is $(b_0 + b_1\theta - A_1t_1 - A_2t_2)/A_3$. After comparing two endpoints, every nonconstant parent event has a normal coordinate of magnitude at least

$$\nu = \frac{1}{82^2} = \frac{1}{6724},$$

because its denominator divides $A_3A_3'$. Its $\theta$-slope is at most 84. All terminal and fixed source faces obey these same, or stronger, bounds. The rational verifier checks these inequalities for every listed endpoint comparison.

Parameter-only identities keep their truth values. The nearest distinct lower parameter contact is $1/25$, at distance $1/600$. Contacts exactly at $1/24$ are only the classified half-face families, which are inactive on the actual child domains by the $1/1008$ and $1/24$ margins. Thus no constant event opens a hidden full-fibre gap.

Take

$$\rho = 10^{-20}, \quad d = 2 \cdot 10^{-23}, \quad e = 10^{-23}, \quad \theta_a = 1/24 - d.$$

Remove parent points in the nonconstant strips $|g(\theta_0)| \leq \rho$. Every face moves by at most

$$84d = 1.68 \cdot 10^{-21} < \rho/4,$$

so all retained endpoint-order signs remain fixed. For ordinary terminal predicates, any change is supported in the old strips $|g(\theta_0)| \leq 2\rho$. In dimensions $2, 3, 4$, inside $[0, 1/2]^r$, each such strip has volume at most $4\rho/\nu$: integrate along a coordinate whose coefficient has magnitude at least $\nu$ and bound the other side lengths by one.

Charge this union of strips separately to five comparator integrals: the discarded positive-parent mass, the three losses, and the positive recovery. Their densities are each at most 200000. The original parent classifications are disjoint, and repeated good predicates are charged within each affected integral; there is no additional Boolean-cell multiplicity. The total cost is at most

$$\begin{aligned} 5 \cdot 200000 \cdot 11\,000\,000 \cdot 4\rho \cdot 6724 &= 0.00295856 \\ &< 0.005 < 0.02. \end{aligned}$$

All of this is a comparator cost. Only whole positive parents and positive terminal failures are discarded in the short sequence.

The small sieve parameters are selected independently of $e$. For example a level exponent $10^{-1000}$, a small-prime exponent $\delta_{\text{sieve}} = 10^{-2000}$, and an internal error parameter $\varepsilon_{\text{sieve}} = \delta_{\text{sieve}}^3 = 10^{-6000}$ preserve the usual source ordering while remaining much smaller than $\rho$ and $e$. The corresponding moment and sieve orders are enormous but fixed. The actual interval exponent is still $\theta_a + e$; using a much smaller internal error parameter does not reduce that interval slack. The fixed fundamental-lemma accuracy is chosen within the remaining positive budget.

We have therefore obtained

$$\Delta = d - e = 10^{-23}, \qquad \theta_* = 1/24 - 10^{-23}.$$

Finally set $c_{\text{new}} = 24/23 - \Delta$. Its strict position above the transfer boundary is exact:

$$c_{\text{new}} - \frac{24}{23 + 24\Delta} = \frac{\Delta(47 - 552\Delta)}{23(23 + 24\Delta)} > 0.$$

The last inequality holds since $\Delta < 47/552$. This proves the explicit exponent claimed in Theorem 1.1 under its stated numerical hypotheses. The calculation bounds a permissible decrement; it does not identify the best one.

# E Quantitative bookkeeping and reproducibility

The arithmetic and geometric parameters are selected first. For $Q = \exp[D(\log\log X)^2]$, one then uses arbitrarily large *fixed* powers $Q^{-R}$ in the mean-square inputs. For every fixed $r, C, \sigma > 0$,

$$(\log X)^C = Q^{o(1)}, \qquad Q^r = X^{o(1)}, \qquad X^{-\sigma} Q^r (\log X)^C \longrightarrow 0.$$

The Perron/mean-square transfer behind the sieve asymptotics has the form

$$\int_X^{2X} |\mathrm{error}(x)|^2 dx \ll X^3 (\log X)^C \max_T \left[ \min(\eta, T^{-1})^2 \int_T^{2T} |F(b+it)|^2 dt \right].$$

An input $\eta^2 Q^{-R}$ and error threshold $\eta X Q^{-r}$ give exceptional measure $O(X(\log X)^C Q^{-R+2r})$. The low-frequency prime-polynomial main term is kept explicitly; at cutoff $Q^K$, its uniform VK error is padded to $Q^{-3K}$. The longer-interval comparison is proved separately in Section 12.

Every factorization depth, divisor order, moment, and variable count is fixed. Thus dyadic or finer rectangular decompositions cost only fixed powers of $\log X$ or $Q$, which can be reserved in advance. Coupled product and ordering cuts can be bracketed by smooth cuts at relative width $Q^{-K}$, with fixed $K$ chosen for the desired accuracy. Conditioning on all but one variable bounds the global boundary count by $O(XQ^{-K}(\log X)^L + X^{1-\delta_0}(\log X)^L)$ for fixed $L$ and some fixed $\delta_0 > 0$. Here is an explicit justification for divisor-bounded coefficients. For fixed $r \geq 1$, $y \asymp N$, and $0 < v \leq N$,

$$\sum_{y < n \leq y+v} \tau_r(n) \ll_r v(\log 2N)^{r-1} + N^{1-1/r}(\log 2N)^{r-1}.$$

Choose a largest factor in each ordered $r$-factorization. The product $q$ of the other factors is $O_r(N^{1-1/r})$; summing $v/q + 1$ with weight $\tau_{r-1}(q)$ gives the bound (the case $r = 1$ is immediate). If the selected variable has length $N \geq X^\delta$, the second term, after the other variables are counted, is at most $X^{1-\delta/r}(\log X)^L$. Thus one may use $\delta_0 = \delta/r$; it need not equal the small-prime exponent itself. Its short-window first moment and Markov's inequality give any required $XQ^{-R}$ exceptional budget after increasing $K$. The resulting Mellin separation has only fixed powers of $Q$ in its norms and frequency truncations. Raise the low-frequency cutoff above those frequency shifts; the uniform prime-polynomial estimate then retains its saving. For the Harman comparison, the cutoff $\exp[(\log X)^{1/3}]$ already dominates every fixed power of $Q$. This fine analytic separation is distinct from the fixed geometric parent strips, whose cost was charged only to comparator integrals. Prime powers in a factor of length at least $X^\delta$ have a power-saving first moment; Markov's inequality consequently also puts their exception inside any fixed $XQ^{-R}$ budget. Relative errors $O(1/\log X)$ in removing prime logarithmic weights, and the fixed fundamental-lemma error, only consume a chosen small portion of the positive main-term margin; they need not be $Q$-small. No theorem with a constant depending on a fixed $B$ is used by replacing $B$ with a growing function of $X$.

The single source file contains standard LaTeX `filecontents` blocks. Compiling it writes the following verification files into the compilation directory:

- `verify_harman_certificates.py` and its two JSON certificates for $\mathcal{H}_1$ and $\mathcal{H}_2$;
- `verify_threefactor_fraction.py`;
- `region_stability_probe.py`;
- `verify_negative_child_coverage.py`;
- `verify_quantitative_sieve_stability.py` and its exact rational report;
- the exploratory `li_III_i0_convention_check.py` and its numerical results.

Run the exact checks with ordinary Python 3:

```
python3 verify_harman_certificates.py
python3 verify_threefactor_fraction.py
python3 region_stability_probe.py
python3 verify_negative_child_coverage.py
python3 verify_quantitative_sieve_stability.py
```

They require only the standard library and use exact rational arithmetic. Expected counts are $70 + 36 = 106$ Harman leaves, 56 classical Hölder identities, 3209 substituted affine forms, and 144 open $U_{02}$ arrangement cells plus 2 $U_{0303}$ cells, with their endpoints, in the original universally quantified child coverings. That coverage check verifies complete affine arrangements, not a sampling grid. The final check verifies the full 496 587-event inventory, the conservative count and normal/slope bounds, the explicit loss budget, and the exact inequalities for $c_{\mathrm{new}}$. The source also writes `mills_chain_5_4.gp`, described in Appendix F. The exploratory integral script additionally uses NumPy and SciPy and is explicitly not a rigorous quadrature certificate. None of these scripts is executed by LaTeX itself.

The exact checks validate the stated finite inequalities and geometry. They do not independently validate the external large-value theorem, all four numerical sieve integrals, or an optimized value of $\Delta$.

# F A finite illustration at exponent $5/4$

The original harmonic-branching manuscript recorded a finite greedy chain starting at $p_1 = 1009$, using PARI/GP's `isprime` for primality proofs [21]. Its first terms are

$$1009,\ 5689,\ 49409,\ 736657,\ 21581503,\ 1470964273,\ 288073066607,\ 211046580457399.$$

The reported 22 terms have digit counts

$$4, 4, 5, 6, 8, 10, 12, 15, 18, 23, 28, 35, 44, 55, 69, 86, 107, 134, 167, 209, 261, 326.$$

This earlier computation is retained as a reproducible illustration; it is not a numerical construction at $c_{\mathrm{new}}$ and is not used in any infinite-existence proof. For $c = 5/4$ every nesting check is an exact integer comparison:

$$m \in J_{5/4}(p) \quad \Longleftrightarrow \quad m^4 \geq p^5 \quad \text{and} \quad (m+1)^4 < (p+1)^5.$$

The source embeds the original GP script as `mills_chain_5_4.gp`. The command

```
gp -q mills_chain_5_4.gp
```

reconstructs the 22 terms, proves each selected term prime, checks every exact nesting inequality, and writes the full chain. The optional decimal floor comparison at the end is a numerical consistency check; the two integer inequalities above are the nesting certificate. The GP primality computation is distinct from the standard-library rational verifiers in Appendix E. No claim that this particular greedy chain continues indefinitely follows from a finite calculation or from the global mass theorem.

## Acknowledgment and revision record

The initial harmonic-branching manuscript acknowledged research development and manuscript preparation with Claude (Anthropic). This revision consolidates that manuscript, the quantitative sieve extension, and the explicit below-1/24 analysis into a single proof sequence. It preserves the external-input status of the numerical sieve margins and includes the exact verification materials used for the new finite inequalities and geometry.

## References

[1] W. H. Mills, A prime-representing function, *Bull. Amer. Math. Soc.* **53** (1947), 604. https://doi.org/10.1090/S0002-9904-1947-08849-2.

[2] B. Chen, V. Gupta, and Y. C. Li, *Large Value Estimates for Dirichlet Polynomials with Characters and Zero Density of Dirichlet L-Functions*, arXiv:2507.08296v2, 27 July 2026. https://arxiv.org/abs/2507.08296v2.

[3] R. Li, *Primes in almost all short intervals III*, author-site manuscript, 13 pages, accessed 9 November 2026. https://runbolicarey.com/assets/downloads/Primes_in_almost_all_short_intervals_III.pdf.

[4] G. Harman, *Prime-Detecting Sieves*, London Mathematical Society Monographs, vol. 33, Princeton University Press, Princeton, NJ, 2007. https://press.princeton.edu/books/hardcover/9780691124377/prime-detecting-sieves-lms-33.

[5] H. Davenport, *Multiplicative Number Theory*, 3rd ed., revised by H. L. Montgomery, Graduate Texts in Math. 74, Springer, 2000.

[6] A. Ivić, *The Riemann Zeta-Function*, Wiley, 1985.

[7] K. Ford, Vinogradov's integral and bounds for the Riemann zeta function, *Proc. London Math. Soc.* (3) **85** (2002), no. 3, 565–633. https://doi.org/10.1112/S0024611502013655.

[8] A. E. Ingham, On the estimation of $N(\sigma, T)$, *Quart. J. Math. Oxford* 11 (1940), 201–202.

[9] M. N. Huxley, On the difference between consecutive primes, *Invent. Math.* 15 (1972), 164–170.

[10] L. Guth and J. Maynard, New large value estimates for Dirichlet polynomials, *Ann. of Math.* (2) **203** (2026), no. 2, 623–675. https://doi.org/10.4007/annals.2026.203.2.6.

[11] B. Saffari, R. C. Vaughan, On the fractional parts of $x/n$ and related sequences II, *Ann. Inst. Fourier* 27 (1977), 1–30.

[12] A. Zaccagnini, Primes in almost all short intervals, *Acta Arith.* 84 (1998), 225–244.

[13] D. Bazzanella, A. Perelli, The exceptional set for the number of primes in short intervals, *J. Number Theory* 80 (2000), 109–124.

[14] A. Gafni, T. Tao, On the number of exceptional intervals to the prime number theorem in short intervals, preprint, arXiv:2505.24017 (2025).

[15] R. Li, *Primes in almost all short intervals*, arXiv:2407.05651v6, 25 September 2025. https://arxiv.org/abs/2407.05651v6.

[16] C. Jia, Almost all short intervals containing prime numbers, *Acta Arith.* **76** (1996), no. 1, 21–84. https://doi.org/10.4064/aa-76-1-21-84.

[17] K. Matomäki, M. Radziwiłł, X. Shao, T. Tao, J. Teräväinen, Higher uniformity of arithmetic functions in short intervals II. Almost all intervals, *Inventiones Mathematicae* 244 (2026), 967–1091. https://doi.org/10.1007/s00222-026-01408-6.

[18] R. Li, *Primes in almost all short intervals II*, Cambridge Open Engage working paper, version 1, 17 October 2025. https://doi.org/10.33774/coe-2025-jrnjl.

[19] K. Matomäki, M. Pandey, J. Pliego, J. Teräväinen, and M. Wang, *Higher order uniformity of the primes and cancellation of the Möbius function in shorter intervals*, arXiv:2610.09567v1, 7 October 2026. https://arxiv.org/abs/2610.09567v1.

[20] N. Watt, Kloosterman sums and a mean value for Dirichlet polynomials, *J. Number Theory* **53** (1995), no. 1, 179–210. https://doi.org/10.1006/jnth.1995.1086.

[21] The PARI Group, *PARI/GP version 2.15.4*, Univ. Bordeaux, 2023, https://pari.math.u-bordeaux.fr/.